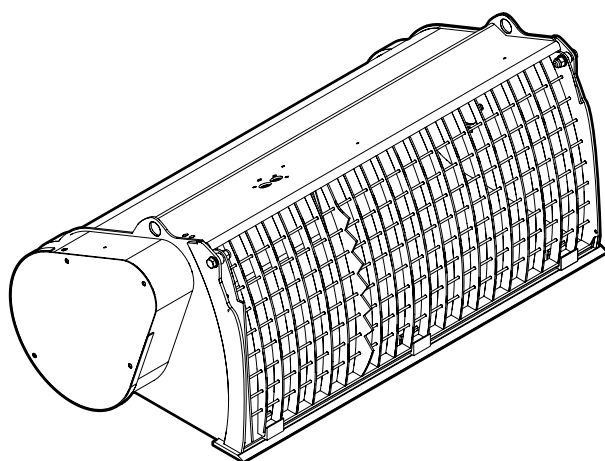


User and Maintenance Manual

Beton mixer bucket

CONDOR

CONDOR SL



**TRANSLATION OF THE
ORIGINAL INSTRUCTIONS**



Edition	11-2022
Printing date	11-2022
Language	EN
Logbook code	MUM 12c290



Attention

Before proceeding with any work on the equipment read this manual in its entirety and ensure you understand the information contained herein.

Keep the manual in a safe place where it is easily accessible for consultation.



U.EMME s.r.l.

Via dell'artigianato 19 - 47015 Modigliana (FC)

Tel. +39 0546 941725 - Fax +39 0546 940050

e-mail: info@uemme.com

www.uemme.com

MANUFACTURER: **U.EMME s.r.l.**
ADDRESS: **Via dell'artigianato 19 -
47015 Modigliana (FC) - ITALY**
DOCUMENT TYPE: **User and Maintenance Manual**
MODEL: **CONDOR - CONDOR SL**
SERIAL NUMBER:
CUSTOMER:
YEAR OF MANUFACTURE:
NAME: **Beton mixer bucket**

Identification of equipment

In order to allow your retailer to help you as quickly as possible, he will need some data concerning the equipment.

Specify these data in this space.

Name

Serial number

Accessories

Address of retailer

Address of
manufacturer

U.EMME s.r.l.

Via dell'artigianato 19 - 47015 Modigliana (FC)

Tel. +39 0546 941725 - Fax +39 0546 940050

e-mail: info@uemme.com

www.uemme.com

Copyright is property of U.Emme S.r.l.

The copy, transcription on other means, translation and use of extracts or parts without the express authorisation of U.Emme S.r.l. is prohibited.

All rights reserved.

The contents of this user and maintenance manual can be edited without prior notice.

With reserve for technical modifications.

WARRANTY

“U.Emme S.r.l.” (hereinafter referred to as “Manufacturer”) guarantees that every new product is free from defects in materials and workmanship when it leaves the plant.

The Manufacturer is committed to replace free of charge unusable pieces returned due to certified material and/or processing defects.

This warranty is valid for 12 (twelve) months from the date of delivery to the first User.

To this purpose, the date indicated on the bill of lading concerning delivery to the first User is valid.

To benefit from the warranty, it is essential that:

- The first User sends a “Warranty Certificate” to the Manufacturer within 10 (ten) days from reception of the machine.

The “Warranty Certificate” must be filled out fully before the User sends it to the Manufacturer.

- The faulty parts must be sent to the Manufacturer’s site to be duly checked, free of charge and accompanied by the identification data carried on the plate applied to the machine.
- The maintenance schedules and timetables provided by the Manufacturer must be respected.

Expenses for transportation of replaced pieces and on-site inspections by our technicians, useful to ascertain the causes of the defect, will be charged to the user.

The examination of defects and their causes is reserved exclusively to the Manufacturer’s personnel or technicians commissioned by the same.

Replaced pieces covered by warranty remain property of the Manufacturer.

The warranty does not cover:

- Components not produced directed by the Manufacturer, for which the relevant manufacturer is responsible.
- Faults deriving from standard wear.
- Faults caused by incorrect use.
- Faults due to carelessness, accidents, inexperience and to use which does not comply with the prescriptions and intended use of the machine.
- Damage deriving from machine downtime and loss of profit.
- Damage produced to persons, objects and animals resulting from a fault.



Attention

This is an interchangeable equipment compliant with Directive 2006/42/EEC.

Should it be applied on a vehicle suitable for driving on the road, in order to comply with the regulations of the highway code, it must be registered on the logbook at the charge and burden of the user.

INSTRUCTIONS UPON DELIVERY

When the machine is delivered, Staff of the Sales Organisation must supply the Customer with the first detailed instructions concerning installation, use and maintenance operations. These instructions are the ones listed hereafter:



IMPORTANT

During the explanation, the Customer must check the box next to the instruction received.

- ☐ Inform the Customer concerning safety regulations which must be complied with; these regulations are indicated on the stickers applied to the machine and in the Owner's manual.
- ☐ Warn the Customer that it is very important to carefully read and understand the Owner's manual before installing or operating the machine. This manual contains the main instructions regarding installation, use and maintenance of the equipment.
- ☐ Instruct the Customer on how to correctly install and detach the equipment.
- ☐ Inform the operator that it is very important to carefully read the Use and Maintenance manual before installing or operating the equipment.
- ☐ Explain to the operator how to use the controls (if any) of the equipment, pointing out the various safety devices and retainers.
- ☐ Instruct the Customer concerning the correct use of the machine and any accessories applied on it.
- ☐ Point out the chapter of the manual about lubrication. It is very important to explain that regular lubrication ensures machine correct operation and long life.
- ☐ With the manual in hand, instruct the Customer about the different maintenance phases, warning him/her of the risks which could arise during this phase.
- ☐ Explain to the Customer how to secure the equipment for road travel. Moreover point out that road travel entails compliance with regulations in force in each individual country as well as those provided in the manual.
- ☐ Help the Customer to fill out the table on page "I" of the manual, as well as the warranty certificate. Once this certificate has been filled out, it must be sent to the Manufacturer.

WARRANTY CERTIFICATE

MACHINE MODEL

FACTORY NUMBER

DATE OF DELIVERY

CUSTOMER

ADDRESS

AREA CODE CITY PROV.

SIGNATURE OF CUSTOMER

RETAILER STAMP

N.B. For acceptance of warranty standards carried on the back.

(Send to Manufacturer within 10 days from the delivery date)



U.EMME s.r.l.
Via dell'Artigianato, 19
47015 MODIGLIANA (FC) - ITALY
Tel. 0546-941725 • Fax 0546-940050
www.uemme.com

"EC" Declaration of conformity

(Annex II A of Directive 2006/42/EC "machinery directive")

The undersigned Molignoni Mario
Hereby declares that the machinery described below:

Generic denomination:	Beton mixer bucket
Function:	Interchangeable equipment
Type/Model:	CONDOR
Serial number:	_____
Manufacturer year:	_____

Complies with the relevant provisions of the "machinery" directive 2006/42/EC and the regulations transposing it into national law.

PERSON AUTHORISED TO COMPILE THE TECHNICAL FILE:

Name: Davide Molignoni

Address: c/o U.EMME s.r.l.

Via dell'Artigianato, 19 - 47015 Modigliana (FC) - Italy.

It is forbidden to set up or install the interchangeable equipment subject of this declaration on supporting machines not complying with laws and/or Directives in force.

Modigliana, li:

Name of the signatory

Molignoni Mario



U.EMME s.r.l.
Via dell'Artigianato, 19
47015 MODIGLIANA (FC) - ITALIA
Tel. 0546-941725 • Fax 0546-940050
www.uemme.com

Origin certificate

no. _____ / _____

This is to certify that - pursuant to articles 76, 108, 114 of the Italian Legislative Decree 30/014/1992 no. 285 - the equipment specified below has been manufactured at the premises of the

- Company: **U.EMME srl**
- Located in: **Via dell'Artigianato, 19
47015 MODIGLIANA (FC) - ITALY**

This is to certify the following exchangeable equipment:

Manufacturer and registered office: **U.EMME srl**
Via dell'Artigianato, 19 - 47015 MODIGLIANA (FC) - ITALY

Type: **BETON MIXER BUCKET**

Type/Model: **CONDOR**

Frame: _____

The equipment mentioned above is factory new.

Modigliana, on:

U.EMME srl
Molignoni Mario

VIEWED BY

Branch office of the Transport Department of.....

Officer name and qualification.

Date

TABLE OF CONTENTS

WARRANTY

INSTRUCTIONS UPON DELIVERY	V
----------------------------------	---

A - GENERAL INFORMATION

LETTER UPON DELIVERY	5
INTRODUCTION	6
HOW TO CONSULT THE MANUAL	6
Topics not covered by the manual	6
Structure of the publication	6
NOTES FOR THE USER	6
Unauthorised modifications	6
User or machine operator	7
MANUFACTURER	7
CHECKING THE SUPPLIED PRODUCT	7
SYMBOLS USED	8
GLOSSARY	8

B - DESCRIPTION

DESCRIPTION OF THE EQUIPMENT	10
IDENTIFICATION DATA PLATE	10
APPLIED SIGNALS	11
Where to apply the signals	13
Models 100-150-200-200SL-250-250SL-300-300SL-350	13
Models 450-600-750-1000	14
MAIN COMPONENTS	15
Models 100-150-200-200SL-250-250SL-300-300SL-350	15
Models 450-600-750-1000	16
OPTIONAL PARTS	17
POSITIONING	18
EQUIPMENT STATUS	19
Work standby	19
Prolonged shut-down	19
Temporary stop	20
Working condition	20
CHARACTERISTICS OF THE WORK CYCLE	20

C - SAFETY

INTENDED USE	21
IMPROPER USE	21
CONFORMITY OF THE PRODUCT	21
FORBIDDEN BEHAVIOURS	22
ENVIRONMENT	22
RESIDUAL RISKS AND DANGERS	23
During the use	23
During maintenance	23

TABLE OF CONTENTS

GENERAL RISKS FOR THE OPERATORS AND EXPOSED PEOPLE.....	23
OPERATING INSTRUCTIONS	24
Placing and working.....	24
Moving, travelling and parking.....	24
VISIBILITY AND LIGHTING	24
ELECTRIC DISCHARGES	25
THE OPERATORS.....	26
Operator of the equipment and driver of the operating machine.....	26
Routine maintenance operator	26
Site or work area safety manager	26
Equipment safety manager.....	26
SAFETY DEVICES	28
 D - TECHNICAL SPECIFICATIONS	
TECHNICAL SPECIFICATIONS.....	29
AIRBORNE NOISE	31
 E - LIFTING AND HANDLING	
FOREWORD	32
SAFETY REQUIREMENTS	32
LIFTING.....	34
Intended hooking points.....	34
HANDLING AND/OR STORAGE	35
PACKING	35
Removing the packing (if present).....	35
DIMENSIONS, WEIGHTS AND LIFTING DIAGRAMS	36
CHECKING THE EQUIPMENT	37
Pre-delivery checks.....	37
FOREWORD	38
 F - INSTALLATION	
INSTALLATION OF THE EQUIPMENT	39
HYDRAULIC CONNECTION	40
ELECTRICAL CONNECTION	41
CHECK THE CORRECT CONNECTIONS.....	42
REMOVING THE EQUIPMENT	43
 G - CONTROLS	
GENERAL WARNINGS REGARDING THE CONTROLS	44
CONTROLS	44
SAFETY DEVICES	45
Transmission protective casing	45
Unloading outlet protection grate.....	46

TABLE OF CONTENTS

H - OPTIONAL

PROTECTION GRATE OPENING KIT.....	47
-----------------------------------	----

I - TROUBLESHOOTING

TROUBLESHOOTING.....	49
----------------------	----

L - USE

GENERAL PRECAUTIONS.....	50
--------------------------	----

INSPECTIONS AND EXAMINATIONS BEFORE STARTING	50
--	----

DURING USE.....	51
Climbing onto and off of the operating machine	52

ROAD CIRCULATION	52
------------------------	----

RECOMMENDATIONS FOR USING THE BUCKET IN COLD CLIMATES.....	53
--	----

WORK CYCLE.....	54
Loading the aggregates.....	54
Loading the cement	55
Water loading and mixing	55
Unloading.....	55
Direct unloading.....	55
Unloading with outlet.....	56
Unloading with flexible hose.....	56

OPENING THE PROTECTION GRATE	57
------------------------------------	----

CLOSING WITH GAS CYLINDERS (OPTIONAL).....	57
--	----

CLOSING USING SCREWS.....	57
---------------------------	----

M - MAINTENANCE

FOREWORD – GENERAL WARNINGS	59
-----------------------------------	----

CONSULTATION OF TECHNICAL DOCUMENTS	61
---	----

SPARE PARTS	61
-------------------	----

BUCKET CONFIGURATION	62
----------------------------	----

SCHEDULED MAINTENANCE	62
Daily checks.....	62
Every 50 operating hours or every week	62
Every 200 hours.....	62
Every year.....	62

MAINTENANCE OPERATIONS	63
Cleaning the beton mixer bucket.....	63

LUBRICATION	64
Preliminary information	64
Lubricating grease comparative table.....	64
Lubricating points.....	65
Supply grease nipples.....	65
Greasing points diagram (Models 100-150-200-200SL-250-250SL-300-300SL-350)	66
Lubrication interval	66
Greasing points diagram (Models 450-600-750-1000)	67

TIGHTENING TORQUE TABLE	67
-------------------------------	----

TABLE OF CONTENTS

ORDERING SPARE PARTS.....	68
PLACING THE MACHINE OUT OF SERVICE FOR A PROLONGED PERIOD OF DISUSE	68
SERVICE LIFE.....	68
CONTROL REGISTER	69
Storage instructions	69
Instructions for the drawing up	69
Authorised persons	69
Storage of the control register	69
Identification of equipment.....	70
Equipment delivery to first owner	72
Transferring the property	73
Mechanism replacement.....	74
Structural elements replacement.....	74
Safety device and relevant component replacement	75
Serious failures and their repairs.....	75
Periodical inspections	76
Inspections	77
General	77
Inspection before use	77
Intervals between inspections	77
Forms for periodical inspections.....	79
 N - SCRAPPING AND DISPOSAL	
WARNING	80
 O - ATTACHMENTS	
WIRING DIAGRAM.....	81
Wiring kit for machines NOT set-up for electric connection	81
Wiring kit for machines set-up for electric connection	82

Letter upon delivery

Dear customer,

We thank you for having chosen the quality U.EMME and we congratulate you for this choice.

Our commitment for product improvement is implemented in the ongoing service that U.EMME offers its customers.

To this end, for any request for technical clarification or interposition on the machine You can call the after-sales service.

U.EMME is capable of carrying out customisations for each customer and therefore can cope with all the issues concerning the equipment.

For any further information you may need, you can dial the telephone numbers here below or you can write to the following e-mail addresses:

telephone +39 0546 941725

fax +39 0546 940050

e-mail info@uemme.com

internet www.uemme.com

Introduction

The manufacturer declines all liability in relation to anomalies due to wrong settings.

Since our products are in continuous evolution, certain details may not exactly correspond to those completing your equipment model.

In such cases, if you are in doubt with regard to the correct operation, consult an authorised service centre. Never proceed by trial and error.

In order to always offer the best product, report any errors or omissions found in the manuals supplied, especially regarding safety situations, advice for improving the equipment and our customer service and whatever else you wish to communicate to this purpose.



Note

When requesting interventions (even by phone) it is important for the Manufacturer to be able to KNOW THE MODEL AND NAME OF THE EQUIPMENT.

For each service request, it is wise to get this information, before calling.

How to consult the manual

Topics not covered by the manual

This publication DOES NOT address the following subjects:

- Extraordinary maintenance.
Such operations must be performed by staff authorised by the Manufacturer.

Structure of the publication

The manual is composed of separate sections with an initial table of contents showing the sequence of titles of the sections, the chapters, and the topics addressed, complete with page numbers.

Page numbering is consecutive.



Note

Italian is declared the official language.



Attention

With “equipment” in this publication we mean the beton mixer bucket.

The means upon which the equipment is installed is called “operating machine”.

Notes for the user

Unauthorised modifications

It is strictly forbidden for anyone to modify the equipment, in its every part, and for whatever reason without the express written authorisation by the Manufacturer.

No agent or representative of the manufacturer is authorised to give instructions which in any way modify the “Instructions for Use”, the safety prescriptions, the guarantee and/or the method of use of the product.

The manufacturer declines all liability in relation to unauthorised modifications and reserves the right to take any actions it deems necessary to protect its interests.

User or machine operator

The user is directly responsible for personal injury or injury to third parties or property damage resulting from:

- improper use of the equipment and of its every part;
- failure to comply with the safety prescriptions and accident-preventing regulations.

Use of the equipment is reserved exclusively to qualified operators.

A qualified operator is construed as a person who has:

- read the "operating instructions" in their entirety;
- understood the concepts expressed in this publication;
- carrying out what was learned by attending the informative meeting in which the retailer or personnel authorised by the manufacturer explains correct and safe use.

It is advisable to ensure that more than one operator attends the training course.



Note

This information course is aimed at presenting the information contained in the "Operating instructions" and to immediately clarify any doubts, substantially improving operator training as requested by regulations in force.

After the initial training course, other personnel can be trained by qualified operators if the owner considers that the qualified operators in question possess the necessary abilities to pass on the skills they have learned.

Manufacturer

The definition of manufacturer refers to the company

U.EMME s.r.l.

Via dell'Artigianato, 19

47015 MODIGLIANA (FC) - ITALY

www.uemme.com

The manufacturer is not responsible of the consequences deriving from an incorrect or improper use of the equipment like, for example:

- methods of use in conflict with the prescribed methods;
- lack of attention in maintenance, in checks during the production process and in checking the efficiency of the tools;
- removal or disabling of active and passive safety devices;
- irresponsible conduct not in compliance with good common practice;
- unauthorised modifications.

Checking the Supplied Product

On receipt of the supplied product check that the delivered material complies with the order and that the "Operating Instructions" are attached.

Upon delivery of the equipment, check for no damages or missing parts. If you notice signs of damage or missing parts contact the manufacturer or LOCAL AGENT.

When the product is received, in the presence of inconsistencies, missing material, or clear signs of damage, inform the manufacturer immediately, write your reservations clearly on the delivery note and immediately send a documented report to the shipping agent's insurance company, complete with photographic evidence of the problem(s).

Symbols used

The following are the symbols used in the manual to recall the reader's attention to the different danger levels when using and servicing the equipment.



Danger

Information or procedure which, if not thoroughly followed, could cause death, serious personal injuries or damage to the equipment. They are often referred to as “residual risks” or, in any event, to danger situations.



Attention

Information or procedure which advises the operator about the best use of the equipment in order to lengthen its life, to avoid damage or loss of programming, to optimise work in compliance with regulations.



Note

Ancillary information.

Glossary

Operator

In compliance with the directive 2006/42/EC, the operator is the person or persons in charge of installing, operating, adjusting, servicing, cleaning, repairing and transporting the equipment.

Danger

Situations or actions that could be the source of possible injury to persons or animals or property damage.

Exposed person (In compliance with directive 2006/42/EC)

Anyone who is entirely or partly inside a hazardous zone.

Risk

Combination of the probability and degree of seriousness of possible injury or harm to health in a dangerous situation.

Hazardous zone (In compliance with directive 2006/42/EC)

Any zone inside and/or close to an operating machine in which the presence of a person constitutes a risk for the safety and health of that person.

Routine maintenance

These are the standard control and maintenance operations on the equipment, which do not require special mechanical knowledge to be carried out.

Such operations include: lubrication, greasing, replacing parts subject to regular wear and correction of loosening caused by use, all of which are to be expected.

These operations can be carried out by the operator in charge of the equipment according to the indications provided in this manual.

Extraordinary maintenance

These are the operations caused by unforeseeable breakage or wear, deriving from particular events during use.

Extraordinary maintenance operations must be compulsorily performed by a specialised operator recognised as such by the machine manufacturer.

Protection

Safety measures that consist in the use of specific technical means, called “protections” (guards, safety devices), to protect persons from potential hazards that cannot be reasonably eliminated or sufficiently restricted by means of design strategies.

Safety device

Electrical or mechanical device that prevents accidents and/or damage to property and personal injury; activation of safety devices may be voluntary when performed by an operator or may be caused automatically by the presence of a potential hazard (opening of a protection or access to a certain area).

Operating machine

Means on which the interchangeable equipment is installed.

Equipment/Interchangeable equipment

The subject of this manual (under Directive 2006/42/EC) installed on the machine defines the intended use.

In this manual, equipment means beton mixer bucket.

Installer

The person who performs the equipment installation on the machine can be identified in:

- The machine manufacturer
- A workshop
- The equipment manufacturer

User and maintenance instructions/User and maintenance manual

Collection of information and instructions necessary to meet the requirements of instructions under Directive 2006/42/EC for the efficient operation and safe conditions of the machine/equipment/interchangeable equipment.

B - DESCRIPTION

Description of the equipment

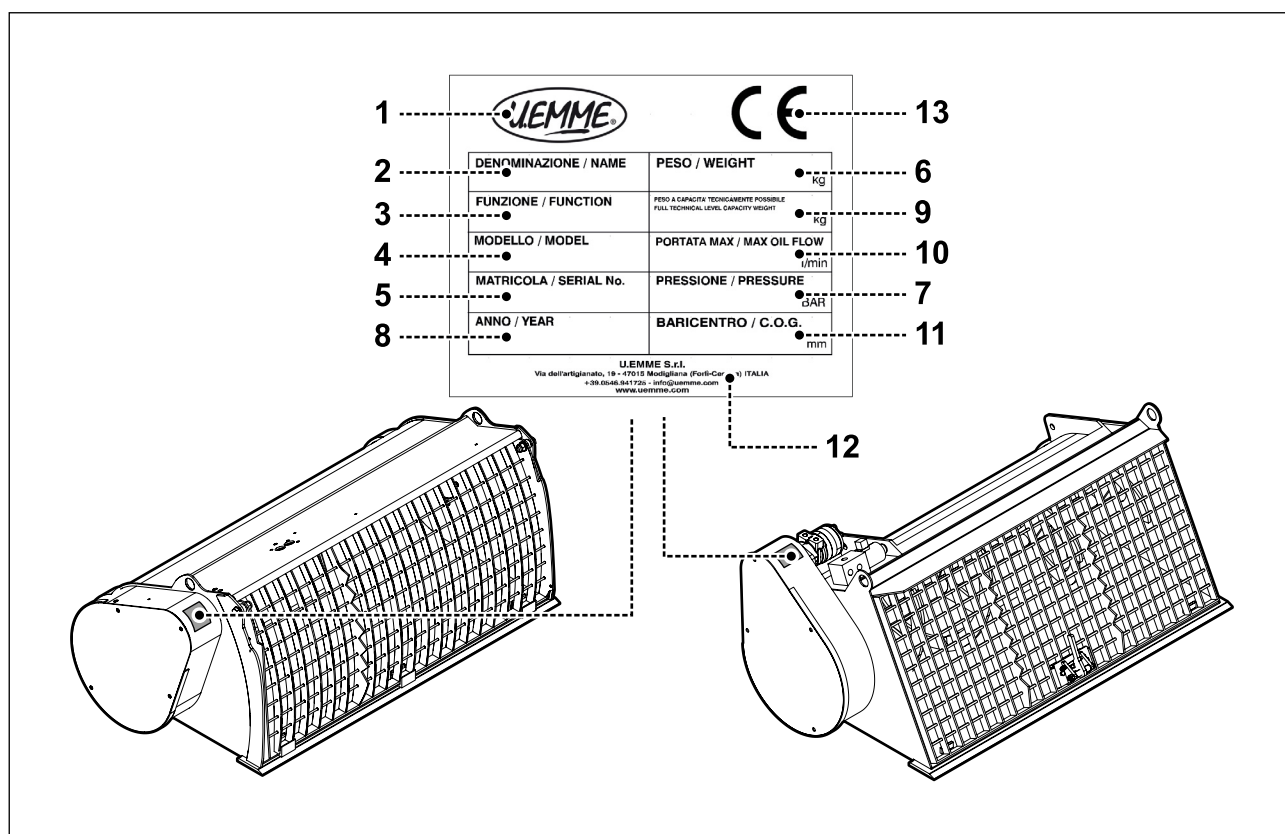
The beton mixer bucket has been designed exclusively to prepare concretes independently from the percentages and granule size of the various materials.

The materials are loaded directly into the bucket.

SL version allows lateral unloading of the concretes.

Identification data plate

Whenever communicating with the manufacturer, always mention the model, general name, serial number and year of manufacture carried on the identification plate.



The identification plate is positioned on the equipment, and carries:

1. Manufacturer logo
2. Generic name
3. Function
4. Model
5. Serial number
6. Weight in Kg
7. Pressure (bar)
8. Year of manufacture
9. Weight capacity technically possible
10. Maximum flow rate (l/min)
11. Centre of Gravity (C.O.G.) (mm)
12. Name and address of the manufacturer
13. EC mark

Applied signals



Stickers are applied on the equipment carrying the safety symbols for a risk-free operation.

Danger

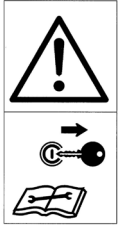
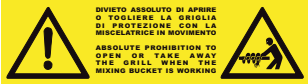
Comply with the specified indications!

To this purpose it is recommended to:

- **Keep the safety stickers clean and easily readable.**
- **Restore those that are missing and replace those that are deteriorated.**

1		Do not approach moving parts Danger of crushing. Keep your lower limbs away from the danger zone. Wait until all moving parts have come to a complete stop.
2		Do not approach moving parts Danger of crushing. Keep your upper limbs away from the danger zone. Wait until all moving parts have come to a complete stop.
3		It is forbidden to stop or pass in the working range of the machine
4		Hooking / lifting points That indicated on the plate is the only point allowed for lifting and handling the bucket with an overhead travelling crane or a crane.
5		Goggles Use protective goggles in order to avoid damage caused by material projection.
6		Helmet Protective helmet to prevent injuries caused by overhanging loads and knocks against structures.
7		Safety footwear with reinforced toecap and non-slip sole Wear accident-prevention footwear to prevent injuries caused by crushing of the feet.
8		Handbook Read and keep the instruction manual with care.
9		Cut-resistant and piercing-resistant gloves Use of protective gloves to avoid cuts, punctures or pricks during operation.

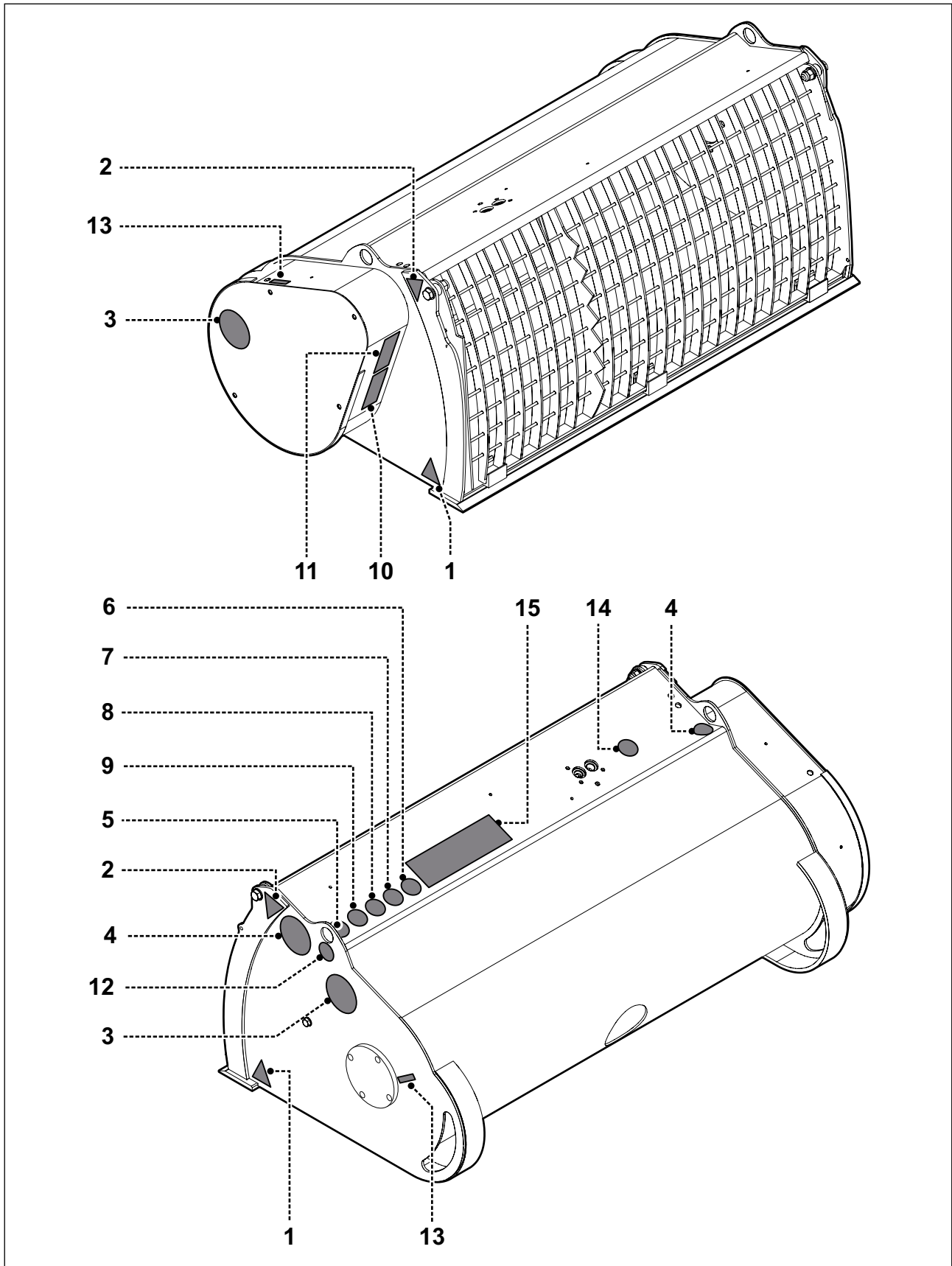
B - DESCRIPTION

10		Read the instructions Read and keep the instruction manual and the safety indications before commissioning.
11		Switch the motor off and remove the key Before performing any maintenance or repair intervention, switch the motor off and remove the key, consult the instruction manual.
12		Lifting of persons is forbidden The use of the bucket to lift or transport operators or personnel is prohibited. Danger of serious injuries.
13		Greasing points
14		It is forbidden to repair, lubricate, adjust, clean machine moving parts
15		It is forbidden to open or remove the grate when the mixer is running.

B - DESCRIPTION

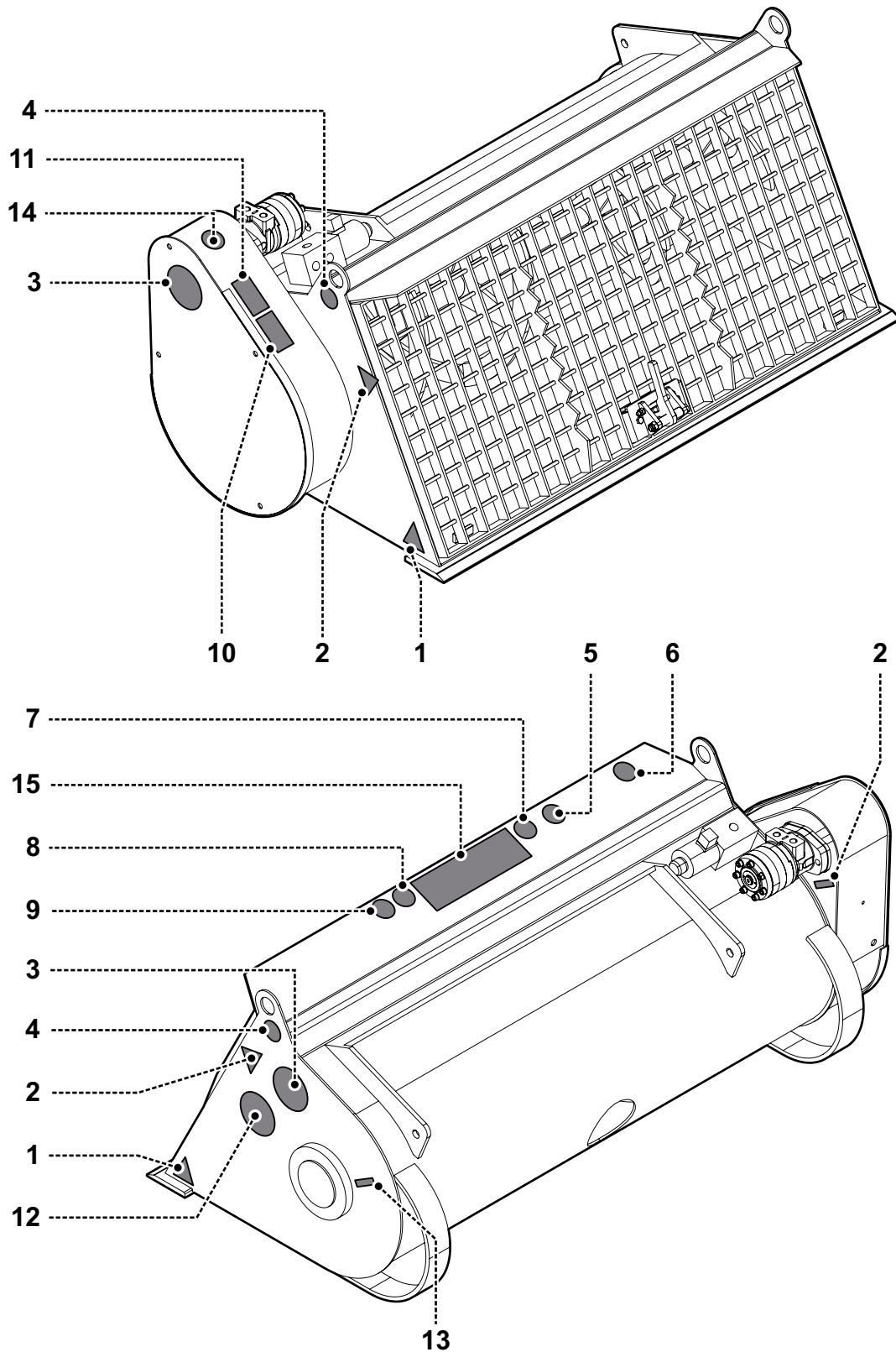
Where to apply the signals

Models 100-150-200-200SL-250-250SL-300-300SL-350



B - DESCRIPTION

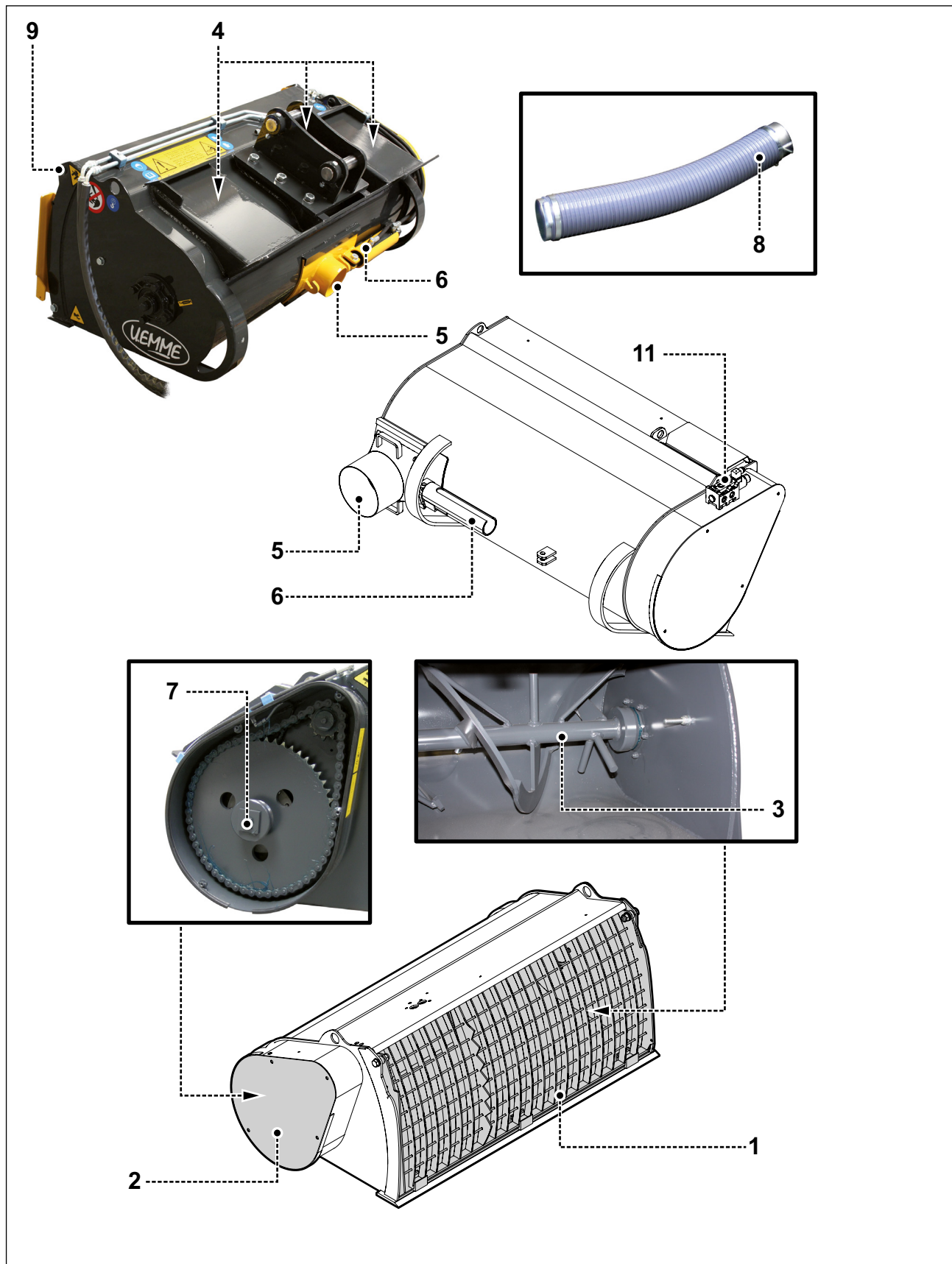
Models 450-600-750-1000



B - DESCRIPTION

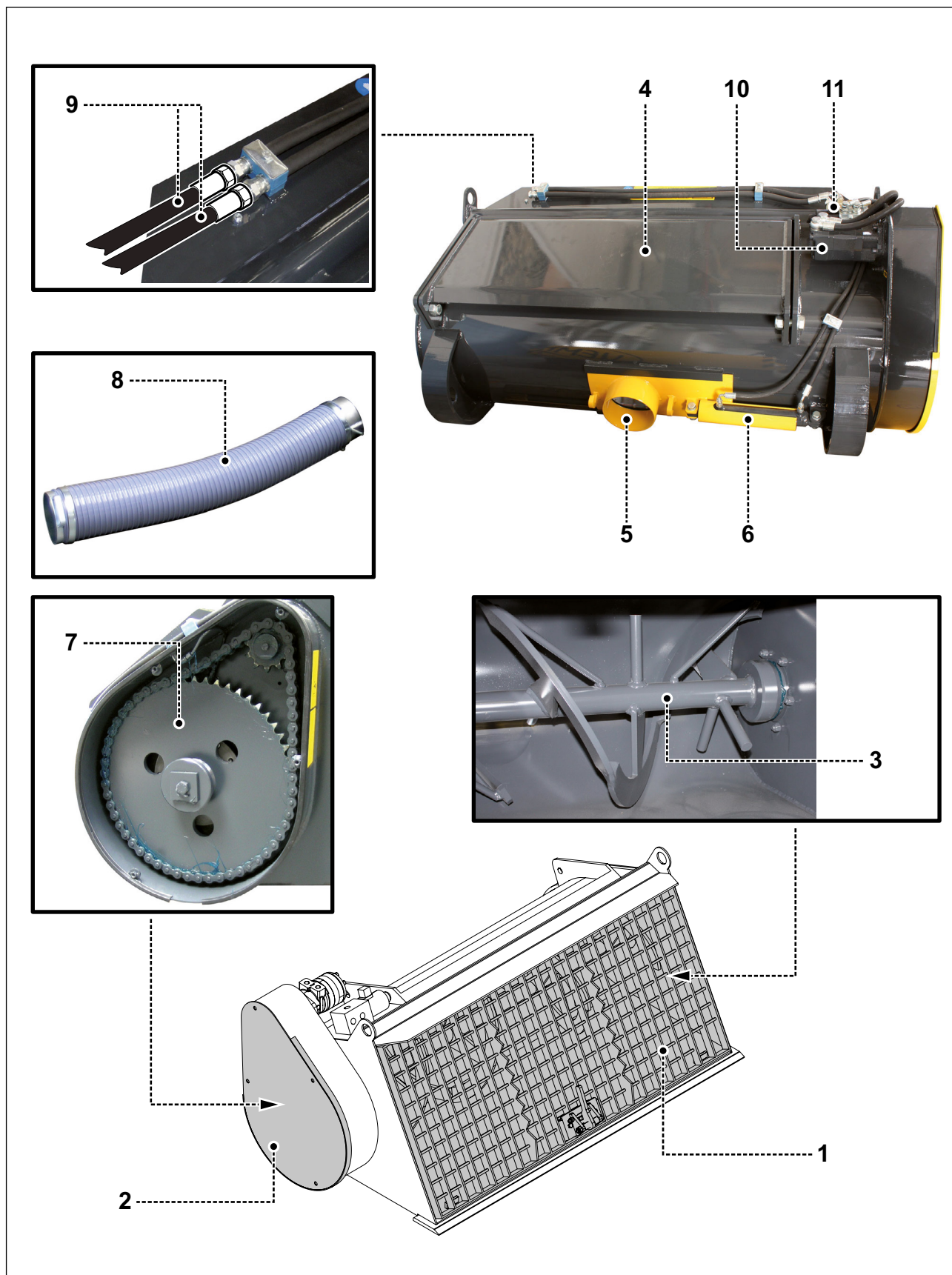
Main components

Models 100-150-200-200SL-250-250SL-300-300SL-350



B - DESCRIPTION

Models 450-600-750-1000



B - DESCRIPTION

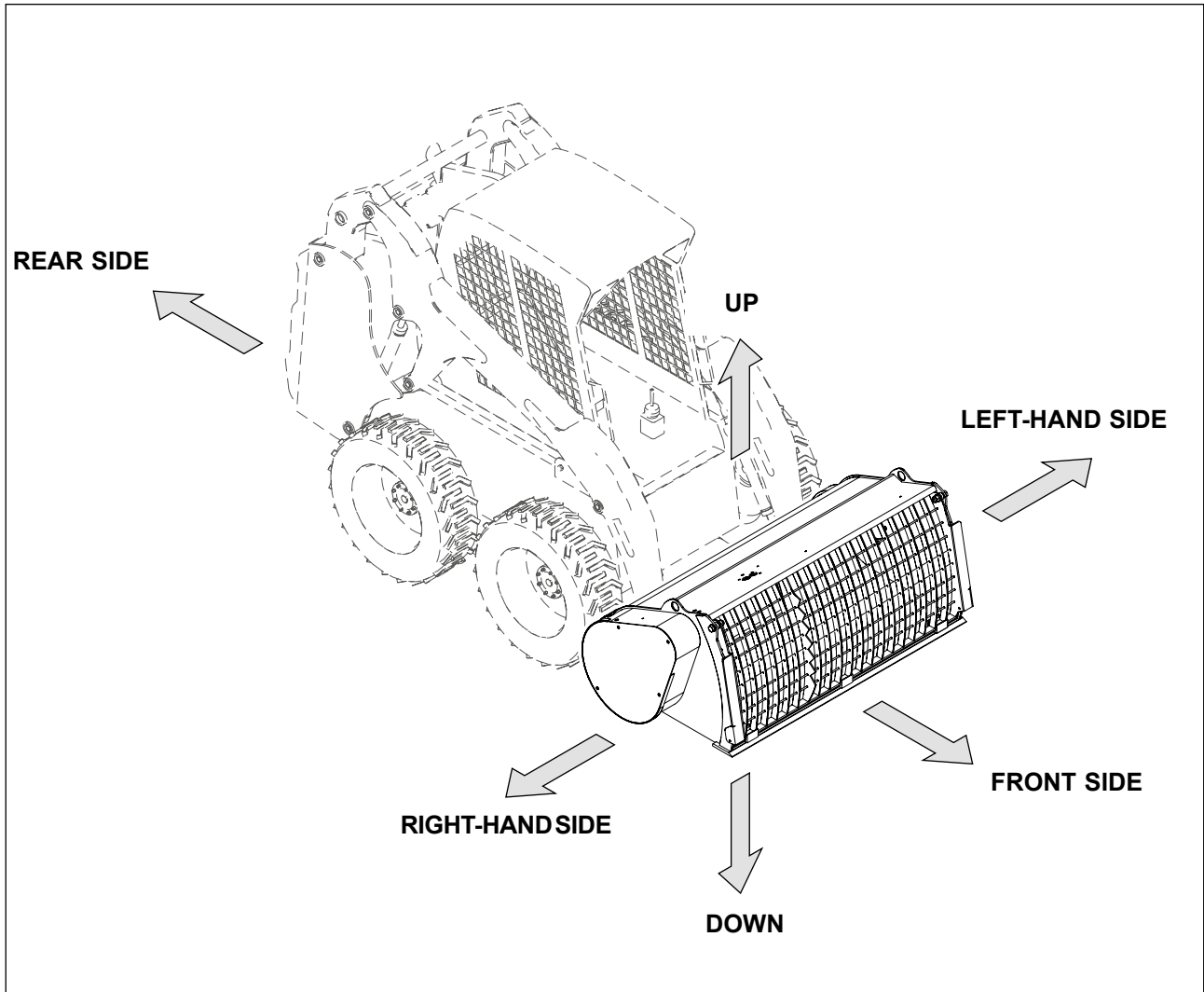
1. Protection grate
2. Transmission casing
3. Auger
4. Attachments to the working machine
5. Outlet for the concrete unload pipe
6. Unload outlet opening/closing system
7. Transmission
8. Concrete unload pipe
9. Quick-coupling hydraulic piping
10. Hydraulic motor
11. Electrical control valve for hydraulic unloading plate opening

Optional parts

- Pressure control valve
- Auger rotation hydraulic block device
- High-pressure and high oil flow motor
- Gas light dampers complete with auger rotation hydraulic block device
- Bolted plate for backhoes
- Quick coupling kit 3/4 S.F.
- 8 or 14 pole connector
- Hardox mouldboard
- Hardox screw shaft (not provided for SL version)
- Cord storage device

B - DESCRIPTION

Positioning



Equipment status

When restarting the machine after any type of stop, make sure no tampering or unauthorised interventions have been carried out (unauthorised release, loosened screws, etc.).

Work standby

By equipment in work standby configuration, we mean those conditions in which the machine is at a standstill for a few hours (e.g.: end of work shift, absence of operator in charge of running and watching over the equipment, lunch break, etc.).

In this case the general conditions must be:

- Empty and clean bucket.
- Equipment resting on the ground in an area suitable for operation and standstill and that guarantees maximum safety, on a flat surface and resting firmly on the ground.
- Equipment connected to the operating machine, with the hydraulic system depressurised (discharge pressure).
- Disabled motor.
- The doors, the panels and all tools provided with lock with padlock or with key, blocked and without keys.
The operator must carry all the keys that activate the operating machine with her/him or store them in a safe place.
- The operator can be temporarily absent from the work area.



Note

If the operator remains in the area, it is not essential to lock doors and panels equipped with locks.

- In the work area there can be other staff, if authorised.
- The work area must be clearly identified.

Prolonged shut-down

When the bucket remains idle for more than 3 days.

In this case the general conditions must be:

- Empty and clean bucket.
- Equipment parked in a suitable place for prolonged standstill (possibly protected) and that guarantees maximum safety, on a flat surface and resting firmly on the ground.
- Equipment resting on the ground on top of wooden supports and NON-yielding ground.
- Equipment disconnected from operating machine.
- The doors, the panels and all tools provided with lock with padlock or with key, blocked and without keys.
The operator must place all the keys which activate the operating machine in a safe place.
- The operator can stay away from the work area.
- In the work area there can be other staff, if authorised.
- The work area must be clearly identified (if work has not been completed).
- Should it be required by the maintenance schedule, all the necessary maintenance operations must be performed.
- Affix safety signs to the machine.



Attention

Install easily understandable and clearly visible signs to prevent admission by unauthorised personnel and indicate that the machine is out of use.

Temporary stop

By temporary stop configuration, we mean those situations in which the equipment has brief standstills.

In this case the general conditions must be as follows:

- The equipment is connected to the operating machine and to power sources.
- The operating machine is on.
- The operator in charge is in the control station.
- In the work area there can be other staff, if authorised.
- The work area is duly identified.

Working condition

By work configuration, we mean those situations in which the equipment is operative and active.

In this case the general conditions must be as follows:

- The equipment is connected to the operating machine and to power sources.
- The operating machine is on.
- The operator is in the control station.
- There must be no other operators in the work area.
- The work area must be clearly identified.

Characteristics of the work cycle

The equipment allows preparing and unloading concrete in two different work phases, as defined in chapter "Intended use".



Danger

All staff must leave the delimited zone and keep a safe distance.

Distance must be calculated considering the most dangerous situation that can occur during an exceptional event.

Intended use

The equipment must be entrusted only to personnel “qualified” for its use by means of adequate “training and information” (at the customer’s charge) and by carefully reading these “Operating instructions” which must be available for the operator before using the equipment.

The equipment must be used by one operator only, except for the unloading operations with flexible hose that require a second operator near the machine.

The equipment has been designed and manufactured to be installed on working machines and used to prepare concretes independent of the percentages and the granule size of the various materials.

The materials are loaded directly into the bucket.

The operating limits of the equipment are subject to the technical data of the various versions (see section “Technical specifications”).

Before commissioning, the installer must perform the technical checks in compliance with the regulations in force.

In particular, the following checks will have to be carried out:

- Visibility
- Controls

Improper use

The equipment must not be used to mix liquids and corrosive materials; use other than the intended one could cause damage to the equipment and constitute a danger for the operators.

It is prohibited to use the equipment outside the limits of use defined in the technical specifications of the different versions (see section “Technical specifications”).



Attention

Use of the equipment under heavy duty conditions (example: bad environmental conditions etc.) entails more frequent checks and/or maintenance operations.

It is forbidden to use the equipment for uses other than that described in chapter “Intended use” and not in compliance with the indications given under section “Safety”.

Conformity of the product

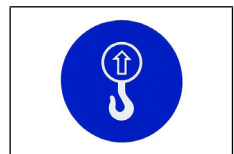
Community Directives

2006/42/EC

Norm of the European Parliament and of the Council regarding the approaching of the laws of the Member-States about machines.

Forbidden behaviours

- It is forbidden to use the equipment for transporting or moving people.
- It is forbidden to attempt to lift up, pull or push loads fixed to the ground.
- It is prohibited to stop in the work area of the equipment.
- It is forbidden to go on working when even a minor structural sagging is detected.
- Intervention is forbidden (adjustments, dismantling, etc.) when the equipment is in motion.
- It is forbidden to approach the equipment wearing unsuitable clothing (e.g. ties, wide, unbuttoned and/or loose clothing, etc.): wear only work wear.
- It is prohibited to use the equipment or not to carry out maintenance interventions without wearing the suitable Personal Protective Equipment.
- It is prohibited to use the equipment in conditions which are not expressly included in its intended use.
- It is prohibited to use equipment lifting points other than those marked.
- It is prohibited to open the grate while the beton mixer bucket is working.



It is prohibited to introduce hands and/or feet inside the unload outlet.



Danger

It is strictly forbidden to load the bucket beyond the maximum level as there is a risk of damaging the components of the drive system (chain, sprocket, crown wheel and motor).

Environment

- It is prohibited to use the equipment in areas with risk of fire.
- It is prohibited to use the equipment in explosive and corrosive environments.
- It is prohibited to use the equipment in closed environments not provided with a suitable air exchange system.
- It is prohibited to operate the equipment at a distance from aerial and underground electrical lines less than the minimum distance allowed the regulations in force in the country of installation.
- It is prohibited to operate with poor lighting.



Danger

If the bucket is applied to a farm tractor, this must be equipped with ROPS (roll over protection structure).

Residual risks and dangers

Below is a list of the risks and dangers that the design and manufacture of the equipment cannot prevent.

During the use

- Danger of crushing and impacts during equipment hooking operation.
- Danger of entrapment and dragging caused by the use of unsuitable clothes.
- Risk of collision with equipment, fixed infrastructures, mobile objects and elements in the work area, if the necessary operating areas are considered and in case handling is careless, approximated and not carried out with attention.
- Risk of collision with resulting crushing dangers for people and/or workers present in the work area.
- Risk for personal safety when two operators must be present in the work area and the operator using the equipment begins to move it before the collaborator has moved away, or moved his limbs away, from the danger zone.
- Risk for personal safety when assembling/disassembling parts, accessories or components and the workers are without the necessary personal protections (safety-helmet, gloves, shoes, etc...).
- Risk of harm to the personnel present in the area when shifting with the bucket lifted over 30 cm from the ground.
- Risk of harm to the personnel in the area when shifting with the bucket provided with teeth, but without the necessary protection.
- Risk of collision with fixed and moving infrastructures along the way (e.g. parked or moving cars, doors, gates, etc...), if the bucket is not positioned properly on the ground after use.
- Risk of crash with fixed and mobile structure along the way, if you do not insert the safety locks in all the mobile parts accessories and devices that need to be locked.

During maintenance

- Danger of material projection during bucket or auger cleaning operations.

Risk of damage to the equipment with consequent danger, if:

- parts or elements of the equipment are replaced with non-original spare parts;
- operations or interventions NOT authorised by the Manufacturer are carried out;
- interventions by shops non-authorised by the manufacturer are carried out.

General risks for the operators and exposed people

If people do NOT stay and/or transit at a safety distance, they can incur:

- Danger of shearing
- Danger of running down
- Danger of crushing
- Danger of impact with consequent falling down

Operating instructions

Placing and working



Attention

Correct and safe use makes it necessary to check the bucket and the proper fastening of the mechanical retainers before beginning operation.

In case of anomalies contact the authorised Service.

Work operations must be performed at a speed suitable to the type of material being moved in order to avoid abrupt and sudden pulls and stress to the structure.



Danger

If too many and too large objects are found along the way, the equipment could be damaged and they could be a source of danger for objects, persons and animals along the work path and not at the due distance.

Keep objects, persons and animals at a safety distance (10 m).

Moving, travelling and parking

Always position the equipment in the middle of the operating machine upon which it is mounted in order not to jeopardise safety during movements or travel.



Attention

Block all the pins with the specific cotter pins before driving or moving the equipment.

If necessary, restore the efficiency of the pins, shear pins and blocks, calling directly the closer Authorised Service Shop.

Equipment and accessories can remain anchored while moving or driving as long as parameters of law relative to maximum allowed clearance dimensions are complied with. Make sure they are blocked/fastened.

However respect the prevailing standards.

Positioning/parking the equipment must not represent a hinder, if there are no prior indications and authorisations, to:

- safety doors, pedestrian crossing and transit ways;
- exits ways in case of danger, doors and private entrances, driveways;
- the visibility of signals (road-signs, warning signs, signal lights etc...).

Visibility and Lighting

Before proceeding with works, check if the minimal standards of visibility have been respected, avoiding overcast and not well-lit environments (fog, smokes etc).

Do not proceed to any operation if reasonable safety and visibility levels are not ensured.

Electric discharges

An electric discharge endangers the life of all exposed personnel; it is therefore recommended to keep the equipment and the load at the distance indicated in the following table.

Un (kV)	D (m)
≤ 1	3
$1 < U_n \leq 30$	3.5
$30 < U_n \leq 132$	5
> 132	7

where:

Un = Nominal voltage

D = Distance

Should there be contact between the equipment and electrical lines or should the distance between them be reduced (due to wind, swaying, etc.) causing dangerous discharges, you must:

- keep personnel and animals present in the area at a distance of at least 10 m from the equipment, from the operating machine or from the load;
- all personnel within a 10 m range must quit the area hopping with their feet together;
- avoid contact with the equipment and operating machine;
- warn personnel in the surrounding area of the looming danger, forbidding them to approach or to touch the equipment and operating machine;
- if your position is free from dangers, do not try to go away, but keep staying in your position waiting for specialised aid, do not touch any object or metallic part different from those you are already in touch with;
- the operators who are in the vehicle cab must remain still in that position, waiting for skilled assistance.

The operators

Every operator has some jobs to do and he must respect them.

The operating figures involved in the use of the equipment can be summarised as follows:

Operator of the equipment and driver of the operating machine

- before using the equipment in work sites or industrial areas, asks the area safety manager detailed information on any dangers present in the area where the operating machine is going to be operated and on dangers that might be created by using the equipment.
- Chooses a suitable place where to operate and park the operating machine with the equipment.
- Delimits the work area to avoid the access of non-authorised people.
- Assesses path to be run with the load and its dangers considering any obstacles on the way.
- Places the equipment in the rest position to transport under maximum safety.
- Makes a visual inspection of the equipment to search for possible faults.
- Makes sure the plates applied on the equipment are in good conditions and well readable.
- Makes sure the equipment is in the correct rest position.
- Drives the vehicle set up with the equipment from one workplace to another under full safety conditions.

Routine maintenance operator

- Checks equipment general conditions, wear of sliding parts, flexible hoses and looks for any leaks.
- Carries out the routine maintenance operations as envisaged and at the intervals specified in this manual.
- Reports unexpected situations to authorised workshops and to the owner of the equipment (such as wear, failure, breakage, etc.).
- Fills out the servicing checks.

Depending on the maintenance operations to be performed, the person assigned to routine maintenance must use the following PPE:

CLOTHES	Use of suitable protective clothing to prevent from being entangled in mechanical parts of the equipment or of the operating machine.
GLOVES	Use of protective gloves to avoid cuts, perforations or picks caused by mechanical parts non-correctly trimmed.
SHOES	Use of safety shoes to avoid risks caused by the falling down of materials or devices used during routine maintenance operations.
GOGGLES	Obligation to protect eyes with goggles or protective screens when operating close to parts of the pressurised hydraulic circuit and/or during machine cleaning operations.

Site or work area safety manager

- Inform the operator in charge of using the equipment about:
 - dangers present in the work area and those that might arise by using the operating machine;
 - possible presence of workers in the danger area (manoeuvre area of operating machine) who, due to particular tasks, cannot leave the work place;
 - possible release of hazardous substances into the air or the ground which could jeopardise the safe execution of work;
 - safety devices compulsory in the working area.

Equipment safety manager

- Informs the equipment operator about dangers which may arise by using it.



Danger

The equipment safety manager must monitor that the equipment is not used improperly; namely that it cannot endanger the health of the operator, of exposed persons, of animals and materials present in the work area.

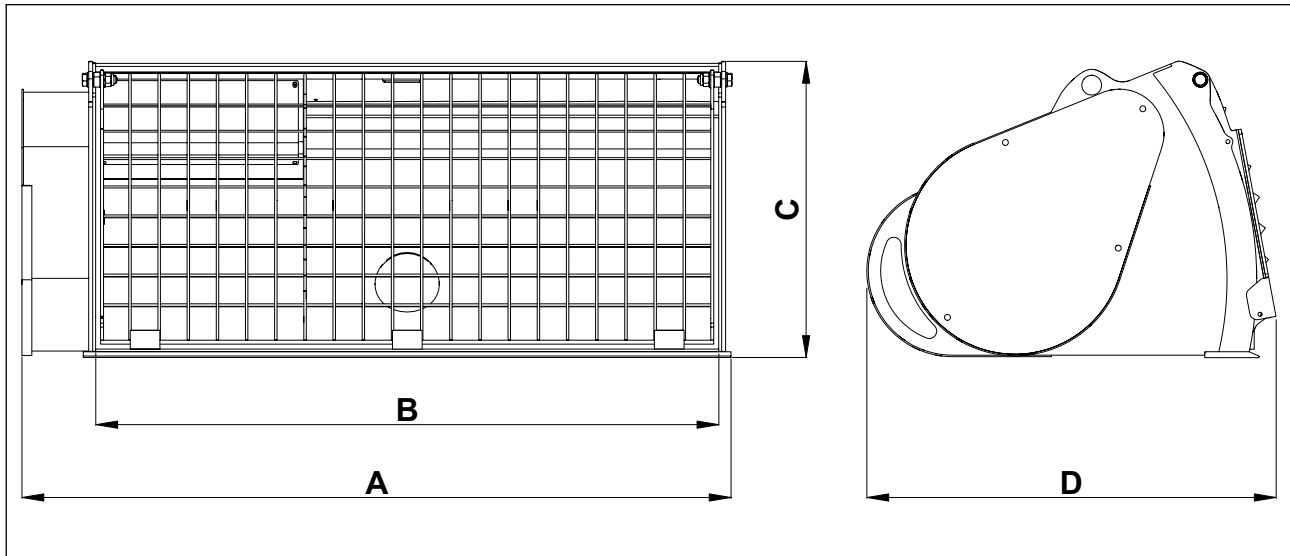
Safety devices

The equipment features the following safety devices:

- Fixed protection guards (casing and/or grates). The machine danger areas are equipped with fixed protection guards. These guards are fixed (for example through screws, nuts, welding) and can only be opened or removed using special tools. This way the access to machine danger zones can occur only intentionally, for example in order to carry out maintenance or repair operations. Such operations are allowed only with the equipment at standstill and with working machine engine off.
- Safety plates and marking. To know the position and meaning of safety plates and markings on the machine, consult the relevant chapter in section B - DESCRIPTION.
- Auger rotation stop device in case of grate assisted opening through gas springs (OPTIONAL).

D - TECHNICAL SPECIFICATIONS

Technical specifications



	100	150	200	200 SL	250	250 SL	300	300 SL
Screw shaft diameter (mm)	310	400	400	400	480	480	480	480
A. Bucket total width (mm)	1110	1170	1490	1490	1490	1490	1575	1575
B. Bucket internal or working width (mm)	940	940	1260	1260	1260	1260	1345	1345
C. Height with universal connection (mm)	630	630	630	630	720	720	720	720
D. Depth with universal connection (mm)	750	750	750	750	820	820	820	820
Unloading hole diameter (mm)	130	130	180	180	180	180	180	180
Weight when empty (kg)	170	255	290	285	345	340	365	360
Technically possible flow rate (l)	100	160	210	210	250	250	270	270
Technically possible overall weight (kg)	411	639	794	781	945	940	1013	1008
Continuous flow rate (l/min.)	60	60	60	60	60	60	60	60
Intermittent flow rate (l/min.)	115	115	115	115	115	115	115	115
Continuous working pressure (bar)	140	140	140	140	140	140	140	140
Intermittent working pressure (bar)	260	260	260	260	260	260	260	260
Continuous torque at motor (Da Nm)	35.7	44.6	44.6	44.6	44.6	44.6	44.6	44.6
Intermittent torque at motor (Da Nm)	66.2	82.8	82.8	82.8	82.8	82.8	82.8	82.8



Danger

The values in the table refer to the standard equipment. To find accurate details, especially concerning weights, refer to the identification data plate applied on the equipment.



Note

Calculations made with 1 l = 2.4 kg ratio.

D - TECHNICAL SPECIFICATIONS

	350	450	600	750	1000
Screw shaft diameter (mm)	480	540	620	620	620
A. Bucket total width (mm)	1700	1760	1760	2070	2370
B. Bucket internal or working width (mm)	1590	1590	1590	1900	2200
C. Height with universal connection (mm)	720	820	950	950	950
D. Depth with universal connection (mm)	820	980	1030	1030	1100
Unloading hole diameter (mm)	180	180	180	180	180
Weight when empty (kg)	420	590	680	760	950
Technically possible flow rate (l)	320	460	600	720	1080
Technically possible overall weight (kg)	1188	1694	2120	2488	3542
Continuous flow rate (l/min.)	80	80	125	200	200
Intermittent flow rate (l/min.)	120	120	150	240	240
Continuous working pressure (bar)	210	210	160	180	180
Intermittent working pressure (bar)	280	280	180	210	210
Continuous torque at motor (Da Nm)	84.3	84.3	106.7	144.5	144.5
Intermittent torque at motor (Da Nm)	112.4	112.4	120.1	168.5	168.5



Danger

The values in the table refer to the standard equipment. To find accurate details, especially concerning weights, refer to the identification data plate applied on the equipment.



Note

Calculations made with 1 l = 2.4 kg ratio.

To couple the equipment - working machine, the following condition must be met:

technically possible overall weight < machine operational rated load ⁽¹⁾ + working machine bucket weight when empty ⁽²⁾



Attention

The calculation cannot replace the execution of the stability tests envisioned before start up.

- ⁽¹⁾ **Machine operational rated load:** usually included in basic machine information, it represents its maximum load, minus the standard bucket mass on which stability loss phenomena can occur even longitudinally. For basic machine and beton mixer bucket correct coupling, this value must be higher than the beton mixer bucket maximum mass in technically possible maximum load conditions. The latter is calculated depending on the beton mixer bucket geometric characteristics, i.e. major operational mixing limitations that can occur when the value indicated by the manufacturer is exceeded, always taking into consideration the more precautionary value for concrete specific weight (2,400 kg/m³). In case of basic machines with variable kinematic (backhoes and industrial trucks with telescopic boom) the comparison must be made taking into consideration the installed safety devices and in particular:

- for the machines equipped with load limitation devices or overload warning devices, the comparison has to be made with the maximum operational net load of the basic machine load diagram;
 - for the machines without limitation and warning devices, the comparison has to be made with the minimum value of the basic machine load diagram.
- ⁽²⁾ only in the cases in which the longitudinal and horizontal distance of the beton mixer bucket centre of gravity from the basic machine supporting structure, under technically possible maximum load conditions, is lower or equal to the distance of the standard bucket centre of gravity.

Airborne noise

The beton mixer bucket is an interchangeable piece of equipment, i.e. cannot operate if not coupled to a thrust vehicle, the noise generated does not consequently exceed the one generated by the working machine.

The user must therefore make sure that the acoustical noise of the vehicle used (farm tractor, operating machine) is within the maximum limit of 80 dB(A).

This measurement must be carried out from the driver's seat of the vehicle.

Should the limit of 80 dB(A) be exceeded, appropriate personal protective equipment must be used (earmuffs, earplugs, etc).

Foreword

The following section carries information regarding handling and moving the equipment.

Comply with all general and specific regulations regarding lifting means and handling and transportation operations, even if not specified in this document.

Before starting working with the machine, it is necessary to have read and learnt the sections of this manual, in particular section C - "Safety".

You must also comply with the specific safety prescriptions in this section to safeguard the operator and the equipment.

Safety requirements

Lifting and handling operations must be carried out with means suitable for the weight to be lifted and handled.



Attention

Operating lifting and handling means must be entrusted to authorised personnel, qualified for their use.

It is the owner's task to find staff qualified for machine lifting and transport, as well as suitable lifting means.

The manufacturer declines all liability related to the use of unsuitable lifting means.

Always use lifting accessories in perfect conditions and suitable for the weight of the equipment or parts to be lifted.

When moving and lifting the equipment, keep the manoeuvre area free from material, objects and persons.

During lifting and handling operations, adopt the necessary precautions in order to prevent the part being handled from falling and/or the means overturning.



Attention

Move the equipment very slowly.

Do not cause blows or jerks and lift the equipment to the minimum indispensable height from the ground.



Attention

For handling and positioning, use lifting means suitable to the weight to be handled (see section "Technical specifications").

All staff, included the operator, will have to keep at a safe distance.

The distance must take into account the hazardous situation which can occur during an unexpected event such as the breakage of a chain or eye-bolt and the consequent overturning of the load.



Attention

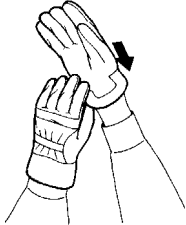
Do not climb onto the equipment for any reason whatsoever.

E - LIFTING AND HANDLING

Before starting lifting operations ensure you are wearing the following personal protective equipment (PPE):



Helmet



Cut-resistant gloves



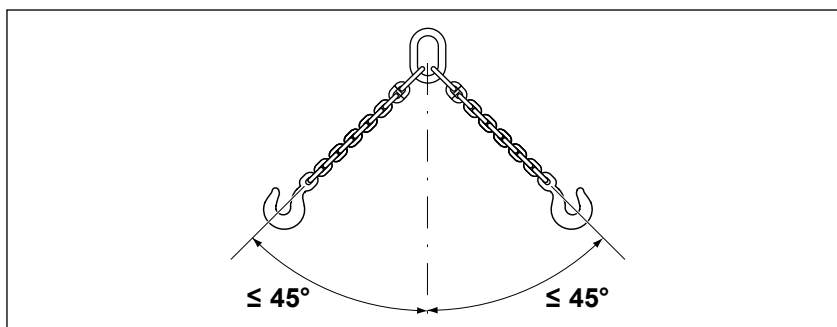
Safety footwear with reinforced toecap and non-slip sole

Lifting

Lifting accessories and harnessing accessories must always be in excellent condition, suitable in relation to the weights to be lifted, in compliance with statutory regulations in force in the country of use, and used in compliance with the manufacturer's instructions and/or the relevant technical standards.

For lifting it is necessary to use chains that allow the machines to be connected to the indicated points (see paragraph "Intended hooking points").

It is crucial that the angle between the chains and the vertical line **does NOT exceed the angle of 45°**.



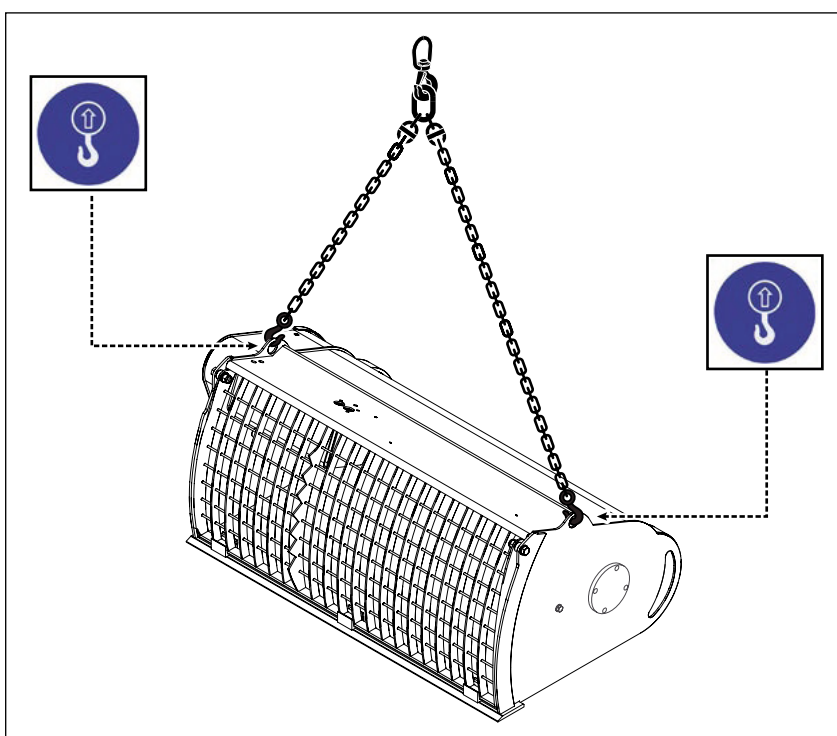
Attention

An angle wider than 45° determines a high stress both on the sling parts and on hooking points, with a high risk of damage and breakage.

Intended hooking points

For lifting and handling of parts, use the lifting points arranged by the Manufacturer.

The lifting points are easily recognisable because highlighted by a pictogram.



It is prohibited to hook the equipment in points other than those envisioned or indicated.



Attention

Damaged pins of the lifting points, hooks or chains are very dangerous because they can break suddenly without prior warning.

Should they be damaged (twisted, loose, corroded etc.) have them scrapped immediately so as to avoid that other operators, unaware of the danger, use them.

Never utilise used accessories if you are not sure they are intact.

Handling and/or storage

Handling and lifting operations must be carried out by specialised companies operating in the machinery transportation sector; these operations can only be carried out in safe conditions with adequate skill and using proper means.

During lifting procedures:

- use the utmost caution;
- keep all persons clear of the area of operations;
- do not allow persons to walk or stand under or in the vicinity of suspended loads;
- lift loads to the indispensable minimum height;
- move the load slowly, without causing blows or jerks,
- keep the manoeuvre area free of materials and clutter;
- use rods long enough for leading the load.

All staff, included the operator, will have to keep at a safe distance.

The distance will have to consider the situation of danger that could arise during an exceptional event, like the break of a chain, a hook and the resulting overturning of the load.

Packing

Removing the packing (if present)

The packing conditions are strictly related to the method of transport and the dimensions of the parts.

While removing packing material protect hands, feet, head, etc. by means of personal protective equipment (PPE) in compliance with standards and in good condition.



Attention

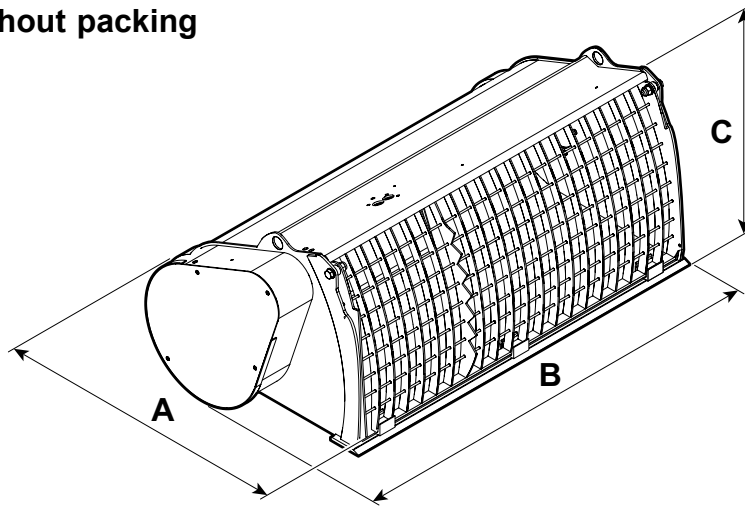
Remove the packing components making sure to separate the materials used to protect the equipment.

Do not pollute the environment with packing, dust, scraps or chemical products. Dispose of them correctly as dictated by local laws.

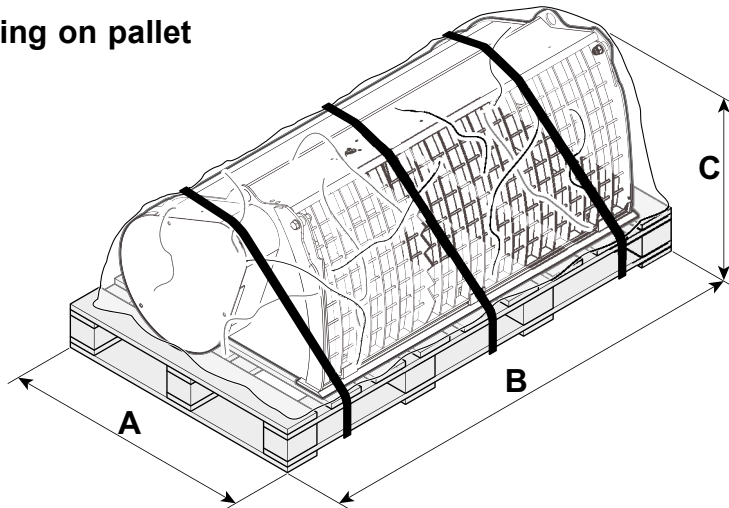
E - LIFTING AND HANDLING

Dimensions, weights and lifting diagrams

without packing



packing on pallet



		100	150	200	200 SL	250	250 SL	300	300 SL
Bucket on pallet without packing	A Width with universal connection (mm)	750	750	750	750	820	820	820	820
	B Length (mm)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)
	C Height with universal connection (mm)	630	630	630	630	720	720	720	720
	Weight when empty (kg)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)
Bucket on pallet with packing	A Width with universal connection (mm)	850	950	950	950	1020	1020	1020	1020
	B Length (mm)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)
	C Height with universal connection (mm)	730	730	730	730	820	820	820	820
	Weight when empty (kg)	(*)	(*)	(*)	(*)	(*)	(*)	(*)	(*)

(*) The choice of the type of packing depends on the model.

E - LIFTING AND HANDLING

		350	450	600	750	1000
Bucket on pallet without packing	A Width with universal connection (mm)	820	980	1030	1030	1100
	B Length (mm)	1700	1760	1760	2070	2370
	C Height with universal connection (mm)	720	820	950	950	950
	Weight when empty (kg)	420	590	680	760	950
Bucket on pallet with packing	A Width with universal connection (mm)	820	980	1030	1030	1100
	B Length (mm)	(*)	(*)	(*)	(*)	(*)
	C Height with universal connection (mm)	820	920	1050	1050	1050
	Weight when empty (kg)	(*)	(*)	(*)	(*)	(*)

(*) The choice of the type of packing depends on the model.

Checking the equipment

Upon delivery, make sure all the parts are included and have not been damaged during transportation; immediately report any defects to your retailer, to the carrier or to the manufacturer.

Pre-delivery checks

The equipment leaves the manufacturing company with all its parts in perfect working order.

In any event, the Retailer, upon delivery to the User, must check the following:

- the hitch for connection to the operating machine,
- all safety stickers indicated in this booklet must be present and legible,
- screws and bolts must be tightened all the way,
- the overall aspect of the equipment must not show imperfections.



Danger

The Dealer must deliver and have the user examine and thoroughly read this manual, which must be kept by the user for the entire life of the equipment.

Foreword



Danger

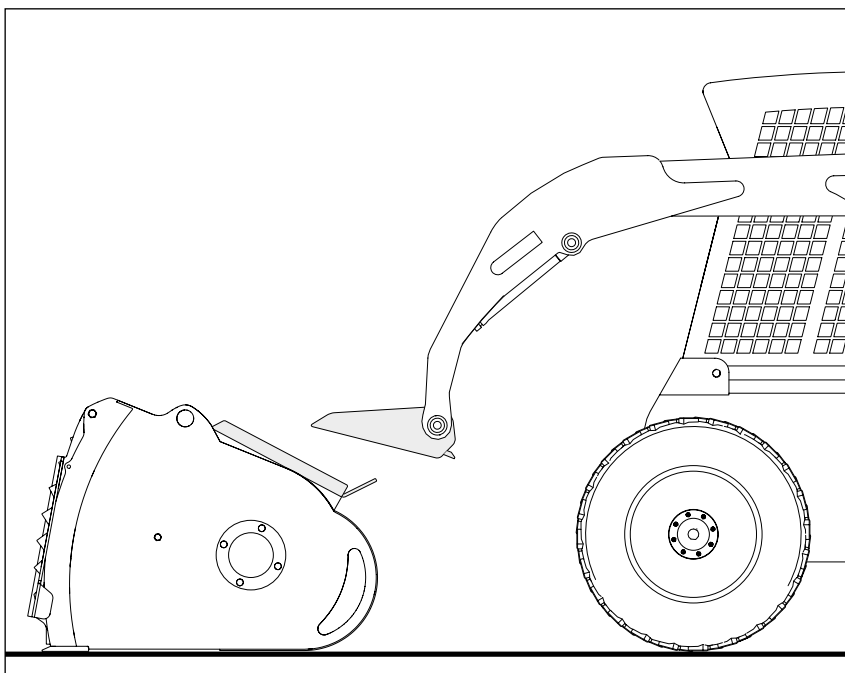
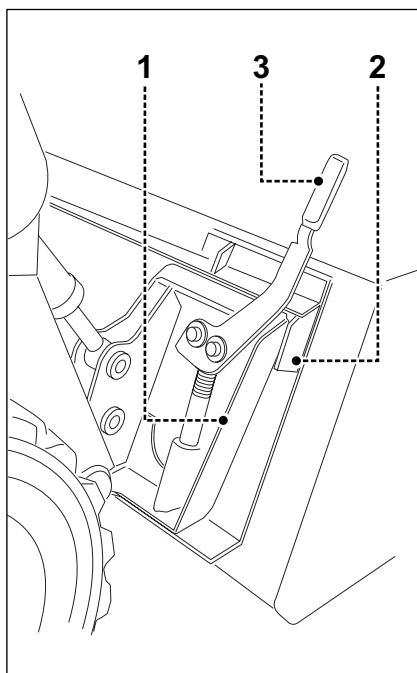
Thoroughly comply with the safety indications stated on the previous pages.

As every operating machine has a tool-holder frame that is different from that present on another one (even if conceptually they can be similar) the information supplied is only general on this type of installation.

More detailed information must be requested to the operating machine Manufacturer.

Installation of the equipment

- Place the bucket on a flat surface that allows performing all working machine approach manoeuvres under full safety conditions.
- Approach with the machine to the rear side of the bucket, making sure to align the working machine tool holder plate with the bucket coupling, stopping at about 50 cm from the same.
- Release the quick coupling devices positioned on the operating machine tool holder plate by working on the relevant levers (or on the relevant lever – see owner's manual of operating machine).
- Lower and tilt the operating machine tool support plate (1) forward.



- Approach with the machine and insert the tool holder plate (1) under the relevant seat (2) of the bucket, paying great attention to perfectly match the coupling seats.



Attention

Perform all connection manoeuvres using the working machine at a very low speed, activating the commands with extreme delicacy and attention.

- Lift slowly and tilt the upper part of the working machine tool holder plate (1) slightly backwards, in a way to make the lower quick couplings of the same match with the respective holes present on the bucket attachment.
- Move the tool holder plate quick couplings device levers (3) to the lock position to fix the bucket on the working machine.
- Check that the wedges or pins of the quick-coupling devices are inserted perfectly inside their seats.

Hydraulic connection



Danger

Before making the connection, make sure that machine motor is at a standstill and the hydraulic system is de-pressurised.

The equipment to be connected is supplied with a range of flexible “Delivery” and “Return” hoses with different quick couplings.

Connect them to the respective couplings of the auxiliary hydraulic system of the operating machine:

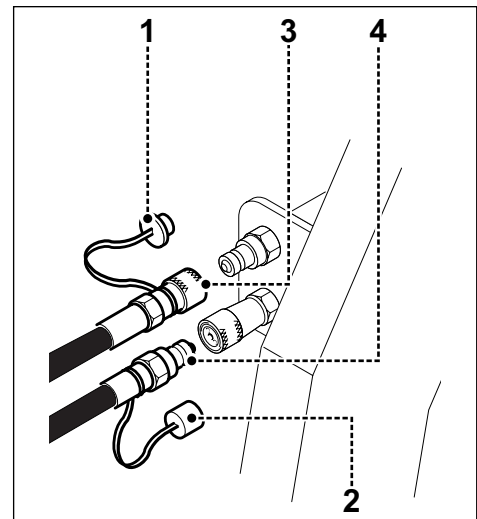
- Remove the protection caps (1-2) from the quick couplings and clean the fittings and respective seats using a cloth in order to eliminate any dust and dirt build-up.



Attention

Clean the bucket and machine quick couplings accurately with the aid of a cloth, before connecting them.

- Connect the quick couplings (3-4) of the flexible hoses to the respective couplings of the operating machine auxiliary system.



The bucket is activated through the command of the working machine aux. system.

Every operating machine has its own type of controls for the auxiliary hydraulic circuit, it is the installer's task to identify the control positions on the machine that respectively allow to obtain:

1 - Mix: Auger clockwise rotation (transmission side)

0 - Auger at standstill

2 - Unload: Auger anti-clockwise rotation (transmission side)

(See section “Control description”).

It is recommended that the controls have a logical sense, e.g. if the aux. hydraulic system control is of the lever type, use the “Forward/Up” positions for the “1-Mix” position and the “Reverse/Down” positions for the “2-Unload” position.

Electrical connection

If the working machine is set up for electrical connection, Uemme supplies the cable for direct connection of the same to the bucket using the control already present on the working machine:

- Connect the connector **A** directly to the coupling of solenoid valve **B**;
- Connect the connector **C** to the working machine.

If the working machine is not set-up for electrical connection, Uemme supplies the relevant cable feeding the solenoid valve and relevant control from machine battery.

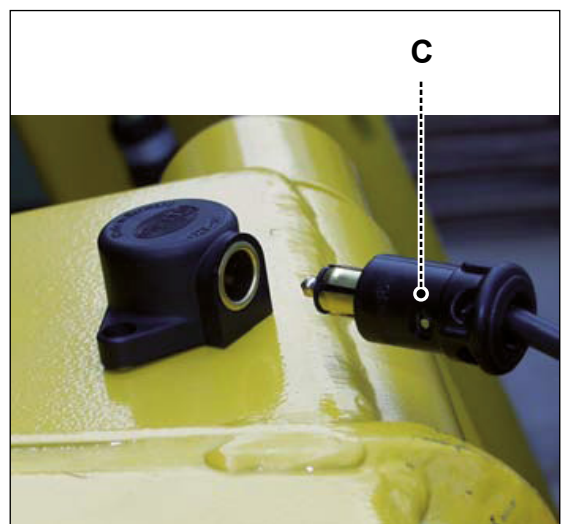
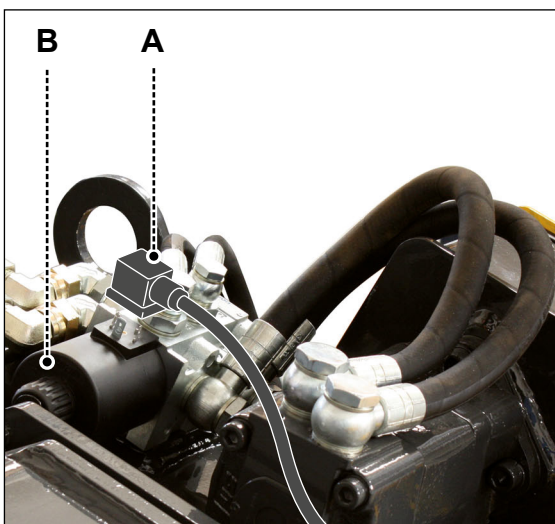


Attention

Have the electric power supply in direct current (12 V) made in an authorised workshop. It is the installer's task to realise the electric system in compliance with the legal provisions and the applicable standards, and to train the operator about correct use.

The bucket does not require any special electric system, however it is recommended to respect the following provisions:

- Identify a polarised socket on the operating machine (the “+ and –” poles must be identified and it must not be possible to invert them), protected upstream by a 10 A fuse, possibly controlled by a switch, which allows to cut-off voltage.
- If the socket is not available on the operating machine, one must be installed with the features indicated previously, making sure to connect the power supply cables directly onto the battery terminals.
- Obtain a bi-polar cable with polarised plug in compliance with the standard of the socket and connect it to the ends of the solenoid valve wires, respecting the polarities (+ and -).
Use a bi-polar cable with suitable length to allow complete movement of the mixer bucket during all work phases (unload included), with suitable section of the wires and isolation.
- Lay the cable along the bucket, making sure to fix it with relevant plastic ties or straps.
Arrange the cable so that it does not hinder operator's movements, paying attention that it does not intercept mobile parts, hinges, passageways, where the risk of it being damaged or being broken exists.



Check the correct connections



Attention

Before starting the bucket, remove any person, animal, tool or object that are within machine working range.

After having connected the flexible hoses, start the machine and check the correct operation of the controls:

- Activate the control of the aux. hydraulic circuit on the “**1-mix**” position and check that the auger turns clockwise (transmission side).
If the controls originates reverse rotation, invert the symbols applied close to the auxiliary circuit control (if this does not create logical problems), or remove the fittings **1** and **2** of the bucket and reverse them.



Danger

Only work on the hydraulic system when the machine has been stopped and the pressure discharged.



Attention

Before reversing the auger direction of rotation, always set the command of machine aux. hydraulic circuit to the “0-auger standstill” position and wait until the auger has stopped. It is prohibited to invert the direction of rotation of the moving auger, so as not to damage the hydraulic motor.

- Activate gate valve opening/closing switch and, using the same auger rotation control, check in which position the gate valve is opened and closed to unload the concrete.
- Let the auger operate briefly in both directions, slightly lifting the bucket from ground, and check there are no oil seepages or leaks.

Removing the equipment

- The bucket must be empty and clean.
- Select a flat surface on which to place the equipment, makes sure it is protected as much as possible from atmospheric agents and from access of unauthorised persons.
- Place the bucket on the ground.
Discharge the pressure in the operating machine auxiliary system as recommended in the relevant sue and maintenance manual, and take the relevant control back to position "0".
- Switch off the operating machine engine, engage the parking brake and remove the key from the dashboard.
- Disconnect the flexible hoses (3-4) of the mixer bucket hydraulic system from the relevant quick couplings (5-6) positioned on the machine, and insert the protection caps on all fittings.
- Disconnect the electric system plug powering the solenoid valve and wind the cable, fixing it to a point on the working machine not interfering with removal operations.
- Release the quick coupling device on the tool holder plate by working on the relevant levers (or on the relevant lever).



Attention

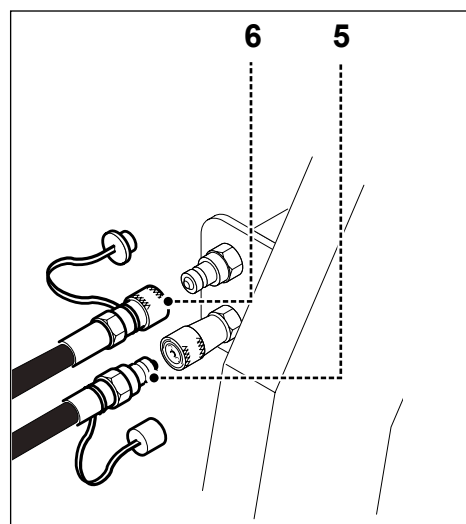
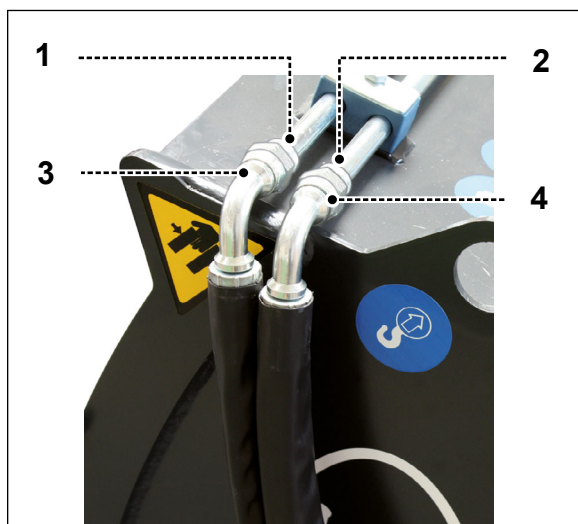
Perform movements slowly and at very low speeds.

- Climb into the driver's seat of the operating machine and start the engine.
- Lift and tilt the machine tool holder plate forward, so as to make it come out of its seat on the bucket.
- Reverse the operating machine.
- Check that the quick couplings on the flexible hoses have protection caps, otherwise connect them to each other.
- If the bucket must remain at a standstill for long periods, it is necessary to place it as indicated in chapter "Prolonged shut-down".



Note

Consult the wiring diagrams in section O - Attachments.



General warnings regarding the controls

- Keep all machine controls in perfect working order at all times.
- Ensure the controls identification plates are always perfectly readable.
- Do not place beverages or other liquid containers on the control console or on other electrical equipment to avoid electric shock hazard in the event of spillage of liquids on electrical parts.



Attention

Before restarting the machine or parts of the machine after an emergency stop, ensure that the causes of the emergency stop have been remedied, and check carefully that there are no persons or obstacles in potentially dangerous areas.

Controls

The beton mixer bucket does not have its own controls, as it is set-up to operate with the aux. hydraulic system of the working machine.

Auger rotation is controlled with the working machine aux. circuit distributor, which can be of the lever type, power or pedal driven, depending on the model of the working machine.

The bucket installer must position a sticker indicating the positions **1-0-2** (or indicate positions **1-0-2**) close to the control of the flow of the aux. circuit of the working machine, to clearly identify the direction of rotation of the auger and the opening or closing of the gate valve to unload the concrete depending on the respective control positions.

The control positions must allow:

1. MIXING AND CLOSING

Clockwise rotation of the auger (view from the transmission side) for concrete mix.

Closing of the gate valve allowing to unload the concrete (after having activated switch **A**).



Attention

Auger rotation for SL version is clockwise (seen from the transmission side) also for unloading the concrete.

0. AUGER AT STANDSTILL

2. UNLOADING AND OPENING

Anti-clockwise rotation of the auger (view from the transmission side) for concrete unload.

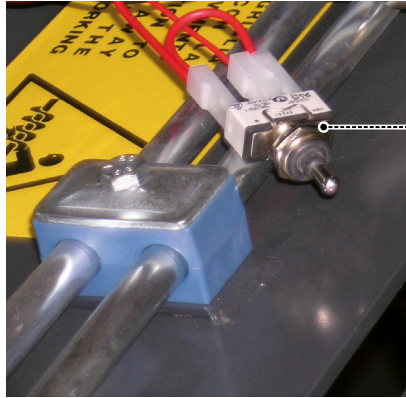
Opening of the gate valve allowing to unload the concrete (after having activated switch **A**).



Attention

Before inverting the direction of rotation of the auger, it is mandatory to take the machine aux. hydraulic circuit control to the "0-Auger standstill" position, and wait for the auger to stop.

It is prohibited to invert the direction of rotation of the moving auger, so as not to damage the hydraulic motor.



A Switch enabling the gate valve operation for concrete unloading

Position	Function
1	Moved to this fixed position, the unload control is always enabled and the auger is at a standstill.
0	In this position, gate valve operation is disabled.
2	In this pulse position, by keeping the position, at every pulse the unload control is enabled and the auger stops just for the time during which the control is activated. Once released, the auger will be activated again.

Safety devices

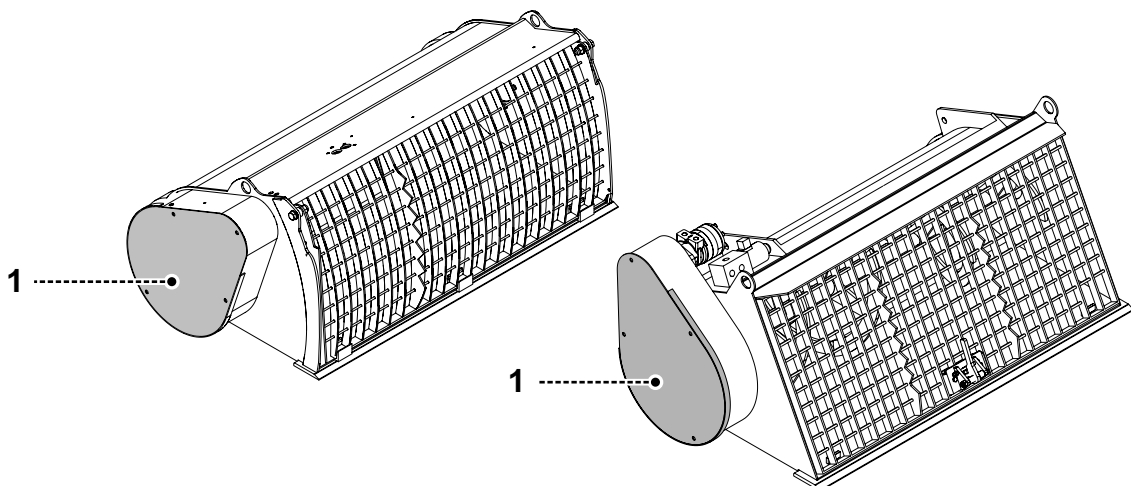
Transmission protective casing

The casing (1) covers the movement transmission parts between the hydraulic motor and the auger (pinions and chains), ensuring a protection against the risk of contact with moving parts during operation.



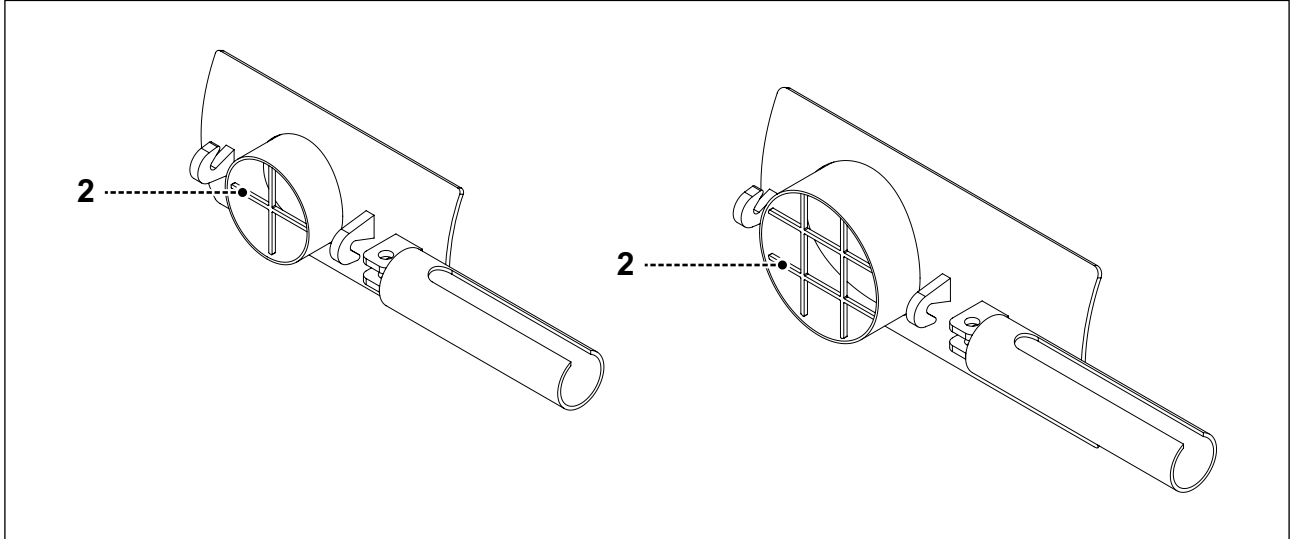
Danger

It is prohibited to use the beton mixer bucket with the casing (1) removed or not correctly fitted.



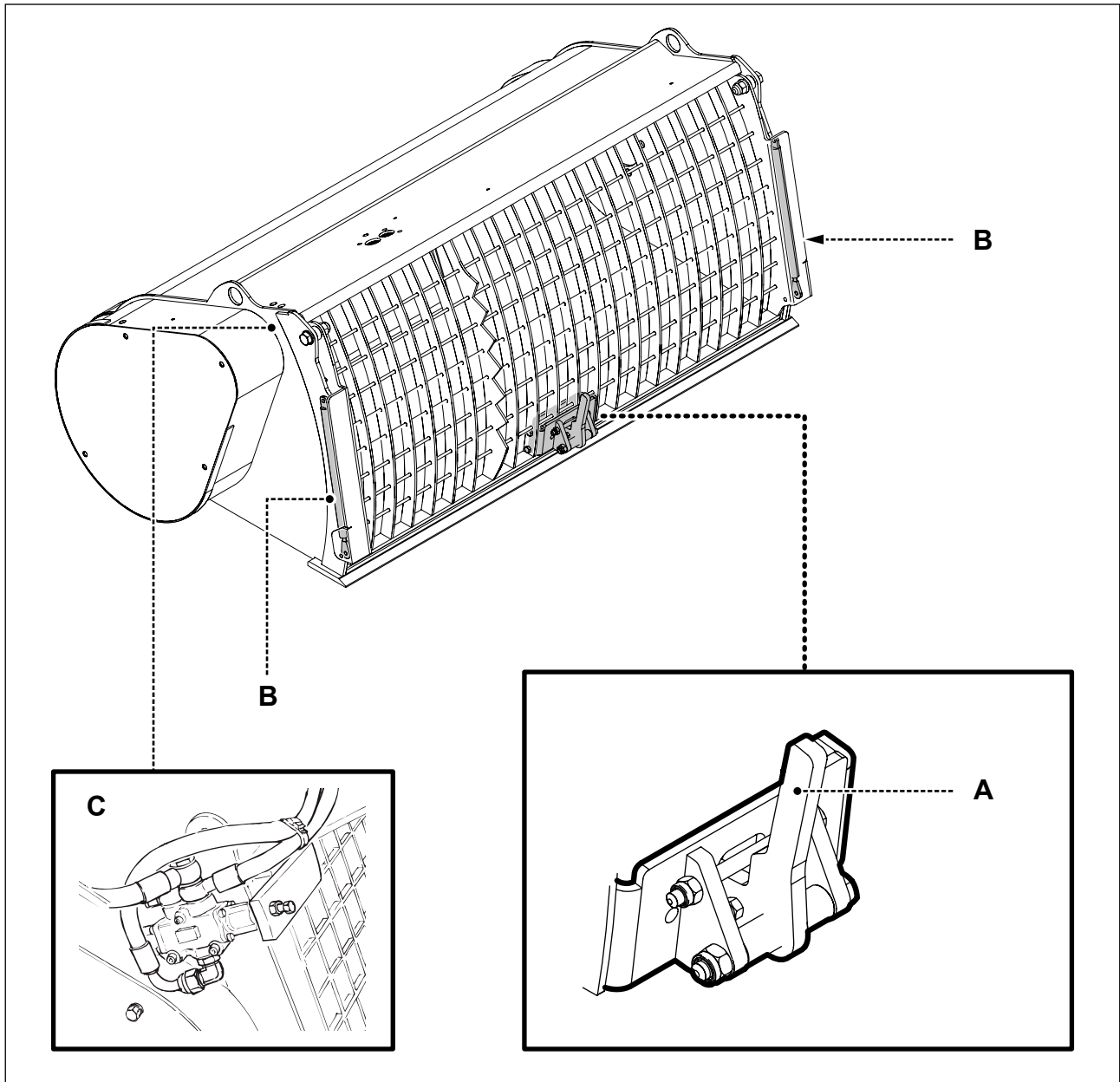
Unloading outlet protection grate

The bucket unloading outlet features a protection grate (2) preventing people from introducing their hands and/or feet inside the bucket.

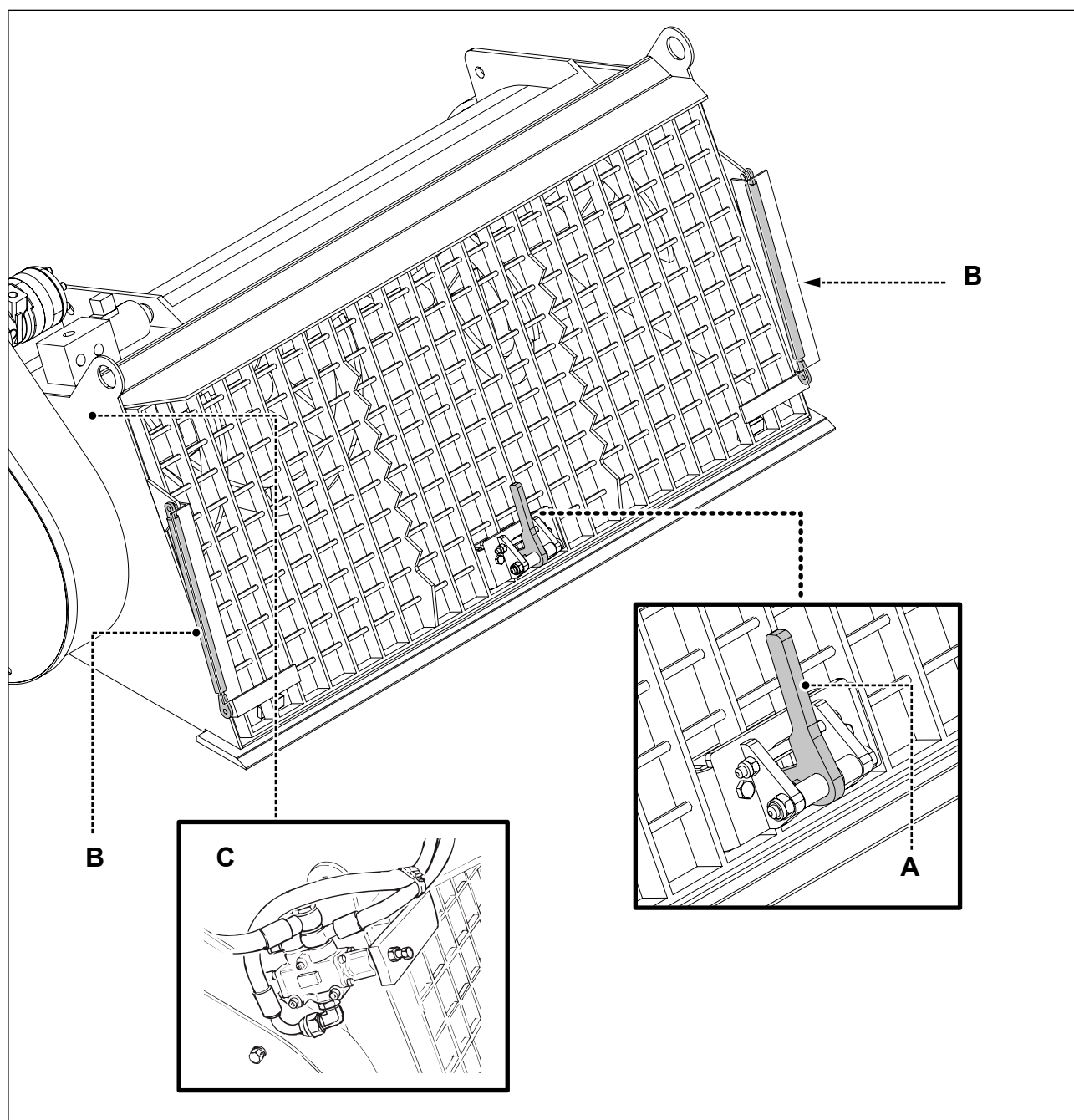


Protection grate opening kit

The kit includes a handle with hook **A**, two gas springs **B** and the auger hydraulic stop device **C**. The kit allows easy opening of the grate, in full safety, since the auger rotation is automatically stopped.



H - OPTIONAL



I - TROUBLESHOOTING

Troubleshooting

Problems	Main causes	Remedies
No mixer auger rotation	The bucket has been loaded over maximum level.	Unload the surplus material from the bucket.
	Oil does not reach the hydraulic motor.	Check the coupling of the hydraulic connections between the bucket and the working machine.
	The transmission chain is broken.	Remove the protection casing and replace the chain.
	The hydraulic motor is broken.	Replace hydraulic motor.
	Open grate in case of equipment with opening kit.	Close the grate.
No unload outlet opening	Oil does not reach the cylinder.	Check the coupling of the hydraulic connections between the bucket and the working machine.
	Current does not reach the valve coil.	Have the points of the concerned electric system checked by an electrician.
	The valve is jammed.	Check that valve piston is not stuck.



Attention

Collect the lubricants contained inside the hydraulic pipes and motor in order to dispose of them separately.

General precautions

Before proceeding with use of the equipment you must have read and understood the previous chapters and in particular section "C - Safety".

If there are still some doubts, refer directly to the Manufacturer's assistance service.

Also read carefully the use and maintenance instructions of the operating machine on which the equipment is installed.

The machine must only be used by qualified staff, which is aware of the position and function of all controls and the instructions given on the various plates.

Uemme guarantees perfect mixing with a load equal to hole capacity (see "Technical specifications")

Inspections and examinations before starting

For a correct use and compliance with the safety parameters, the following inspections must be carried out before every operation:

- Correct hitching of equipment to operating machine.



Attention

All blocking joint pins must have their cotters ensuring a correct coupling.

- Check integrity of the hydraulic hoses and any oil seepage from hoses and fittings.
- Inspection of the carpentry and detection of possible fissures and cracks, paying special attention to the welded areas (discontinuities and cracks in paint that could anticipate damages to the structure).
- Integrity and legibility of diagrams, symbols and warnings present on the equipment.
- Safety devices correct operation.

Also check:

- that the various parts intended to be greased have been lubricated.
- that protections have been correctly fitted;
- the conditions of concrete unload gate valve;
- the conditions and proper tightening of the various parts;
- that the envisioned daily maintenance operations have been carried out.



Attention

When restarting work after a stop or after having moved away from the equipment temporarily, make sure that the devices which were set before going away have not been modified or that the equipment has not undergone vandalism or tampering.



Danger

Before starting any operation, make sure that nobody is present within machine working area.

Pre-warn of the start of manoeuvres with relevant signals and make persons and animals present in the machine work area move away.

Check the integrity of the control indications.

Check the hydraulic oil level of the operating machine (see relevant Use and Maintenance Manual).



Attention

If breakage, even partial, of bucket components are detected, immediately contact a workshop authorised by the Manufacturer, in order to perform the necessary repair operations before using the machine.



Attention

It is strictly forbidden to perform makeshift repairs in order to start work.

During use

- It is strictly forbidden to lean on moving parts.
- Do not use the equipment for any purpose other than that intended by the Manufacturer.
- Always check in a precautionary manner that the operation of the equipment and of its units, even auxiliary, do not trigger dangerous situations for persons, objects or animals.
- Get individual protecting systems foreseen by the prevailing accident prevention standards.
- In case of risk forecast, stop the operations and start working again only after all risk conditions have been eliminated.
- Before beginning work, inspect the operating area to pre-emptively ascertain the presence of obstacles.
Avoid hitting against obstacles since they could damage the equipment or jeopardise the stability of the operating machine.



Attention

Pre-emptively check for the presence of pipes or manhole covers, as well as ground loading capacity.

- Put the bucket in transport position, even for brief journeys (see "Road circulation").
- Always climb up or downhill with the bucket facing the hill, keeping it as low as possible.
- Never exceed the maximum gradients allowed by the Manufacturer of the operating machine.



Danger

During the job, always keep the machine manoeuvre area under control and prevent persons and animals from approaching.

Stop the operating machine immediately if Safety Standards are not respected.

- Make sure you have full vision of the operating area.
Make sure that lighting is sufficient in the event of night-time work.
If necessary, indicate the work area using the relevant signs.
- Do not work close to open excavations, without respecting the minimum safety distances.
- Only activate the controls from the operating machine driver position.
- Avoid sudden movements, act smoothly and gradually on the controls.
- Do not perform adjustments or interventions on the bucket while it is working.
- Do not open the protection grate while the bucket is working.
- Do not introduce your hands and/or feet inside the unloading outlet.

Climbing onto and off of the operating machine

- To climb up and down, use the handles on the side of the ROPS structure uprights or the steps present on the operating machine.
- Access the machine cab, sit on the control seat and fasten the seat belt or lower the protection bar, which usually also enables the machine controls.
Pay attention not to slip or trip over flexible hoses or obstacles present.
- When climbing down from the machine, stop the engine, engage the parking brake, remove the ignition key from the dashboard and follow the reverse procedure.



Danger

Do not climb onto or off of the machine using systems other than those prescribed by the Manufacturer (such as the wheels of the machine or other supports as a step).

Always use the slip-proof steps or platforms envisioned by the Manufacturer of the machine and bucket.

Road circulation

When driving on the road, strictly comply with regulations in force regarding road transportation in the country where you are operating.

- Make sure that oil does not leak on the road.
- Turn the equipment in such a way for the driver of the operating machine to see well.
- Do not circulate on roads if the application is not type-approved in the ways envisioned and recorded in the operating machine log book.
- If the machine-bucket coupling is type-approved to circulate on roads, respect the following provisions if they do not oppose those indicated on the type-approval certificate (or similar document):
 - Yellow rotary light present on the operating machine.
 - Put the bucket in transport position, vertical lifted from the ground by 30÷40 cm.
 - The bucket must be empty.
 - The protection grate must be closed.
 - The working machine aux. hydraulic system must be disabled and with the relevant control set on "**0-auger standstill**".
 - Block any pins present with the relevant safety cotters.
 - Position the signal devices on the front side of the bucket (profiles or reflecting panels for projecting loads, retro reflectors, lights, etc.), which are requested by the standards in force or indicated on the type-approval certificate.
- Circulating on roads or on public land, as long as it is allowed, the regulations prescribed by the Highway Code in force in the country of use must be respected.
- During road circulation, pay great attention close to inhabited areas, crossroads, bridges, subways, level crossings etc.



Attention

If the bucket is not type-approved for road circulation, it must be disconnected from the machine and loaded onto a suitable means of transport.

Recommendations for using the bucket in cold climates

If the equipment must be used in cold climates, with the temperature between **-10°C** and **+5°C**, it is very important:

- To check that the hydraulic oil contained in the machine is suitable for these temperatures (see lubricant correspondence table if envisioned for the operating machine).
- Before starting the bucket, start the working machine and make the pump work at idle for about 5 minutes, so as to take hydraulic oil to operating temperature.
- Activate the aux. hydraulic system and let the auger rotate for a few minutes, always empty.

Work cycle

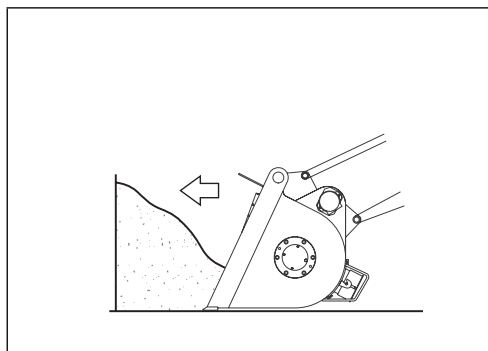
Use of the beton mixer bucket has 3 main phases:

- loading phase (aggregates, cement, water);
- mixing phase;
- unloading phase.

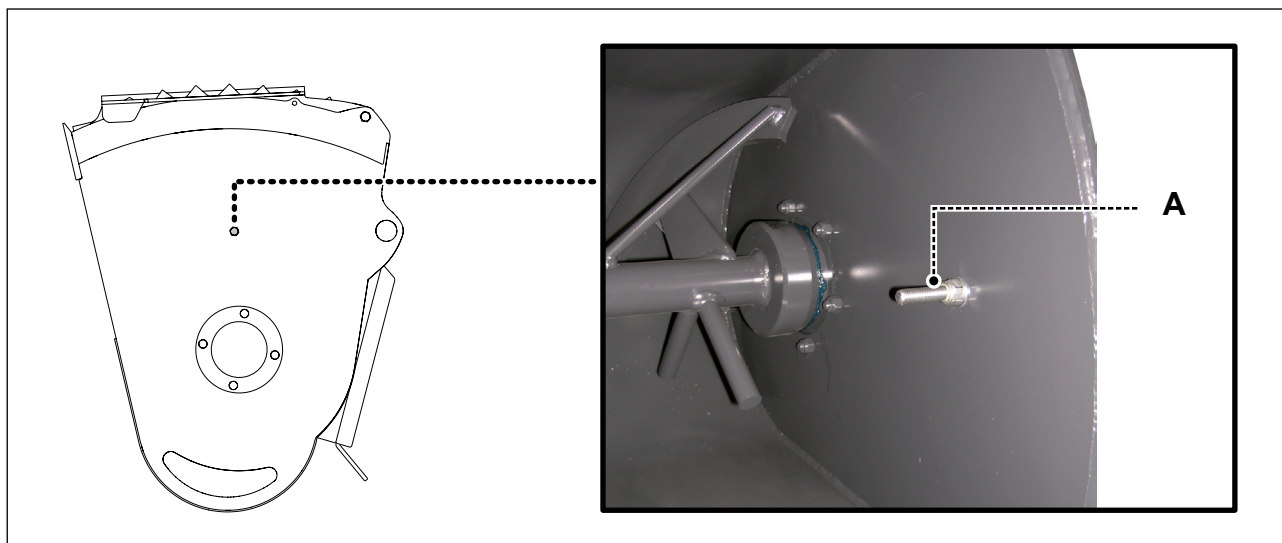
Loading the aggregates

- Start the motor and move towards the mound of earth.
- Perform this operation in a straight manner so as not to stress the bucket transversally.
- Retract the bucket and position it with the loading outlet upwards.

In this phase the auger is at a standstill and the outlet is closed.



Check that the level of aggregates does not exceed the maximum level indicated by screw **A** (see also "Technical specifications").



Danger

It is strictly forbidden to load the bucket beyond the maximum level as there is a risk of damaging the components of the drive system (chain, sprocket, crown wheel and motor).

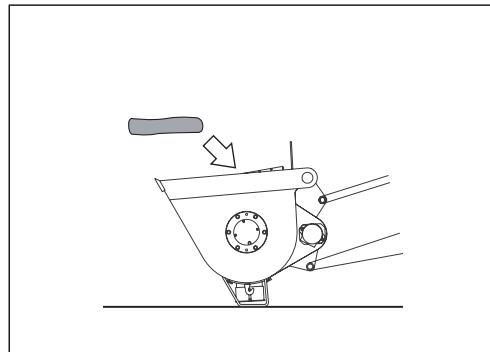
Loading the cement

If you do not have an automatic cement distribution plant, the operator loads the cement manually from the bags. In this case, stop the working machine motor with the bucket on the ground, the loading outlet upwards and the grate closed.



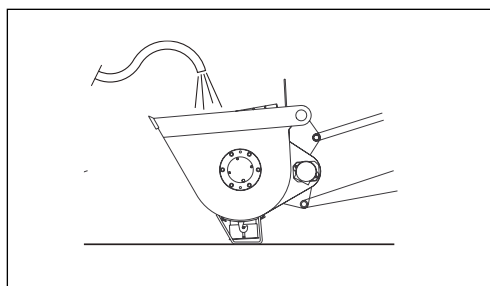
Danger

Use protective masks during the loading phase.



Water loading and mixing

- Start the motor and activate the aux. hydraulic circuit, setting the relevant control on "1-mix". The auger must rotate clockwise (view from transmission side).
- Fill water keeping a safe distance from the grate, preventing the water loading pipe from coming into contact with the moving auger.
- Should auger block during mixing, immediately enable auger reverse movement control, until it is released.
- Invert the movement again to re-start working.



Danger

It is prohibited to remove the protection grate during the concrete preparation phases.



Danger

Use protective goggles during water and mix loading phases.

Unloading



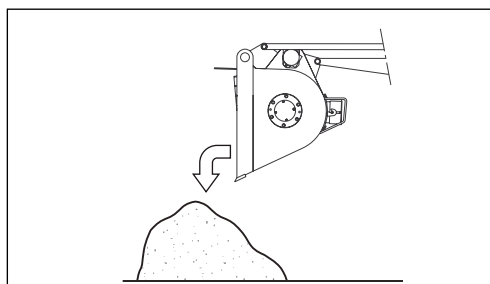
Danger

Check that the concrete does not exceed the maximum recommended level so as not to create dangerous unstable situations with loads and movements at high heights.

Depending on the job being performed, it is possible to unload in different ways.

Direct unloading

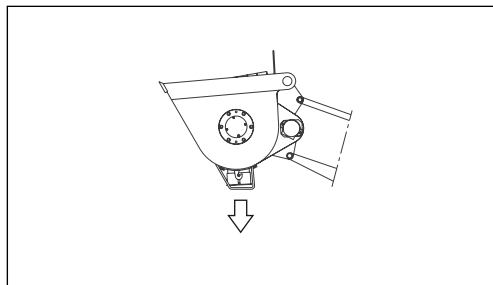
Turn the bucket until concrete comes out of the loading outlet.



Unloading with outlet

When in the unload position (with loading outlet upwards), open the rear outlet.

The auger must rotate during unloading so as to make flow and consequent unloading of the concrete towards the centre of the bucket easier.



Unloading with flexible hose

- Stop the working machine motor with the bucket at approx. 1.5 m from the ground.
- Place on the operating machine lifting cylinders the relevant safety retainers so as to avoid equipment accidental falling down.

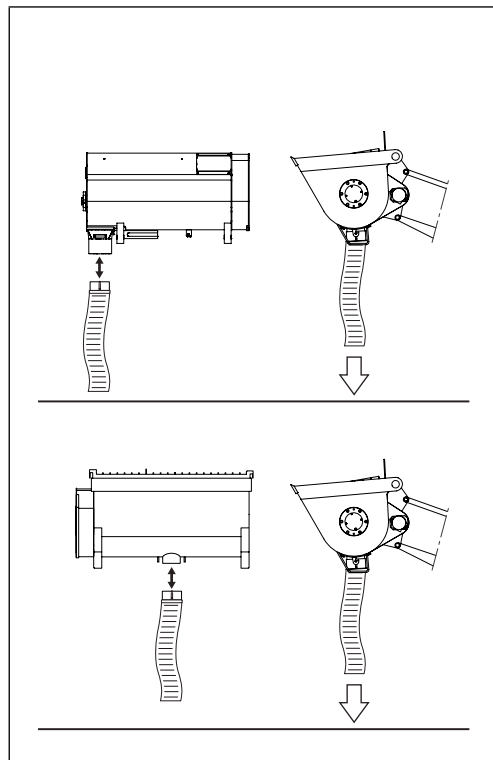


Danger

Fit the concrete unload pipe only after installing the safety retainers on the lifting cylinders of the working machine and keeping the operator body outside the bucket clearance.

Make sure that the working machine and the bucket are in a steady position.

- Fit the unload pipe by coupling it to the outlet support and secure it in place it with a 90° rotation.
- Start the working machine and activate concrete unloading after having enabled it with the corresponding switch.
- Once unloading operations are completed, remove the flexible hose from the bucket and then the safety retainers from the lifting cylinders.



Attention

It is necessary to clean the bucket at the end of every production cycle in order to free it from any residues.

Opening the protection grate

For deeper cleaning or maintenance operations, the protection grate can be opened:



Danger

Before opening or removing the protection grate, stop the working machine engine and remove the keys from the ignition switch.

It is prohibited to open or remove the protection grate when the mixer is working.

Closing with gas cylinders (optional)

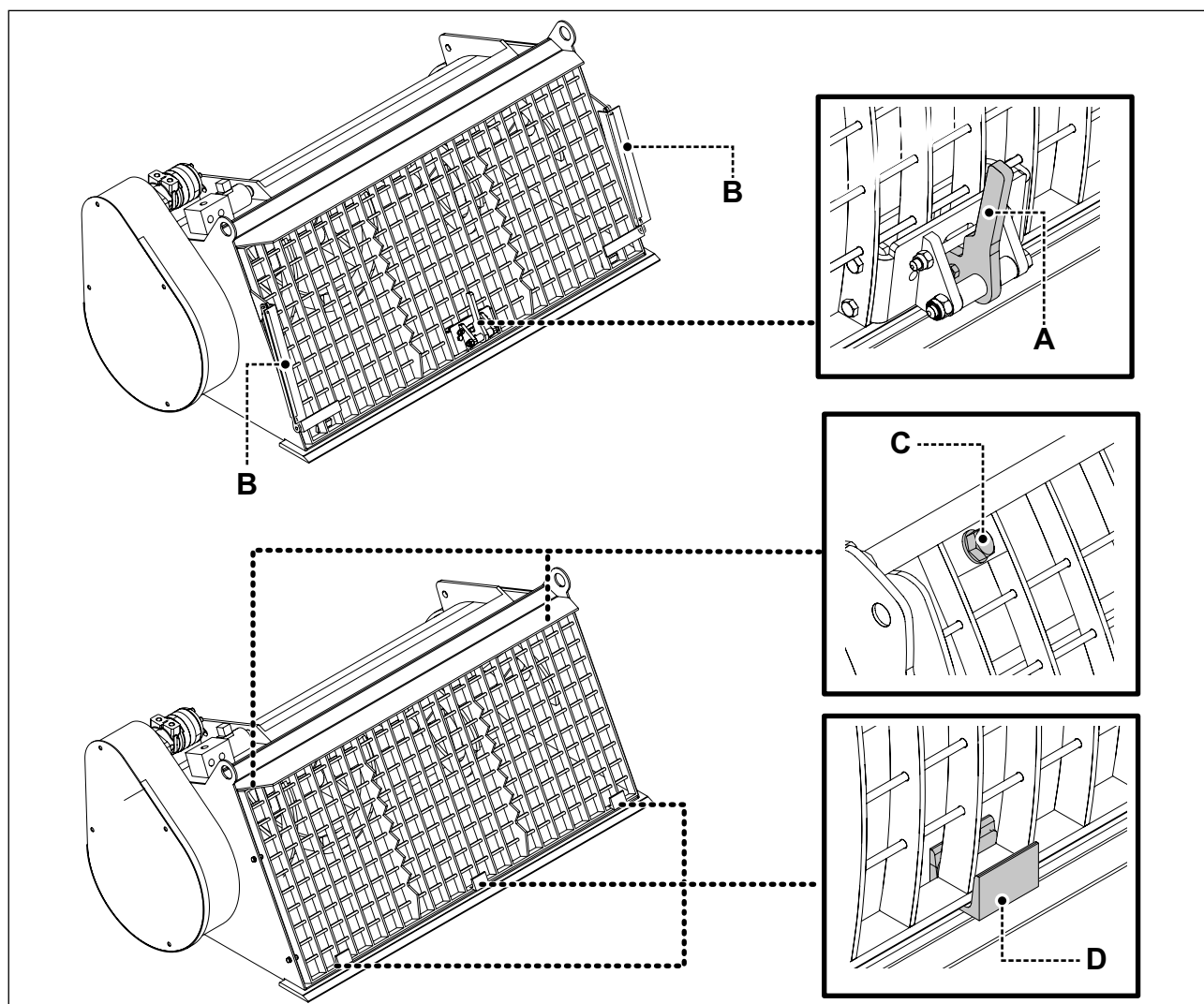
Pull the lever **A** to release lock, the gas cylinders **B** lift the grate and keep it in open position.

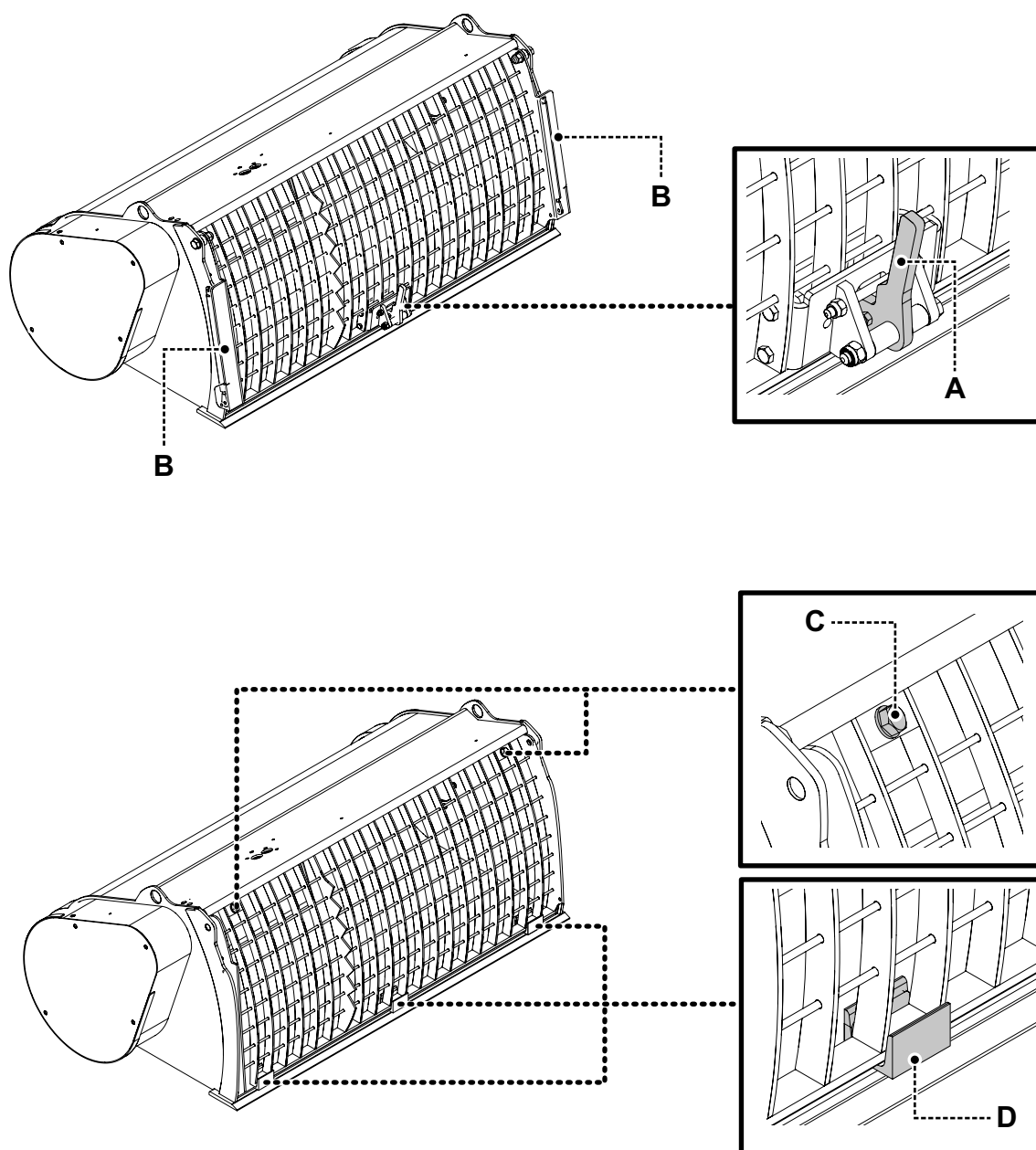
To close the grate, push it until hooking bucket lock.

Closing using screws

Loosen screws **C** and remove the grate from the lower clamps **D** (if present).

To close the grate again, position it on the three lower clamps **D** (if present) and secure it in place with screws **C**.





Foreword – General warnings



Attention

All the operations described in this section must be carried out by maintenance operators, skilled and personnel trained for that purpose.



Danger

Even if not expressly instructed, it is imperative that the machine is cut off from all power supplies before any maintenance work (part replacement, repairs, cleaning, lubrication, etc.) is performed.

Make sure that no other persons are present close to the equipment during use and maintenance.



Attention

Wear suitable Personal Protective Equipment before carrying out maintenance operations.

Maintenance operations must be carried out at least at the recommended intervals, although the exact frequency depends on the conditions of use of the machine.

During maintenance, repair, cleaning and adjustment operations, indicate machine standstill in a visible manner using a sign positioned in the driver's seat or on the distributor indicating "WORK IN PROGRESS".



Attention

Before starting the machine up again, correctly refit and tighten all the parts which have been removed (in particular fixed and moving covers and safety components).

Any operation NOT listed below must be performed by specialised personnel authorised by the Manufacturer.



Danger

Read the "Safety" section of this manual in its entirety before starting work.

- Methodical and accurate maintenance reduces the risk of damage or accidents and preserves the bucket through time.
- The main causes of accidents are due to:
 - lack of oil and grease;
 - dirt accumulated on the various units or components;
 - safety devices out of order; hydraulic system failure (worn flexible hoses, loose fittings; etc.);
 - errors made during maintenance.
- If some operations require the operating machine arms to be lifted, these must be suitably locked in position with relevant stays.

- Never postpone maintenance or repair operations.
- Always follow maintenance and repair procedures requesting previous authorisation.
- Stop the engine and make sure that the pressure has been discharged from all systems before removing casing, guards and covers.
See the machine Use and Maintenance Manual.
- Use the relevant handholds and steps to climb on and off of the machine.
- Do not wear rings, watches, jewellery, open and loose clothing, e.g. ties, ripped items, scarves, unbuttoned jackets or overalls with open zips that may become entangled in moving parts.
It is advised to use approved items for accident-prevention purposes: helmets, slip-proof shoes, protective gloves, anti-drumming hearing protection, reflecting jackets, anti-dust masks, breathing apparatus, safety goggles, when the job requires the same.
Consult the employer for the safety prescriptions in force and use of the accident-prevention devices.
- Do not go under the bucket when this is just lifted.
- If it is absolutely necessary to lift the bucket disconnected from the working machine, use suitable means (see "Lifting and handling"). After having lifted it, always insert a stand or tail prop, leaving the lifting means always in traction.
- Never introduce your head, body, limbs, hands, feet or fingers into a shearing area, without guards, without first having locked tightly any parts that can move.
- Never align holes or slots using your fingers; use a suitable centring device.
- When using compressed air to clean parts, use protective goggles with side visors.
Limit pressure to a maximum of 2 bar.
- Never use petrol or solvents or other flammable liquids for cleaning.
Use authorised, non-flammable and non-toxic solvents available on the market.
- Do not lubricate, repair or adjust the bucket when it is working, unless expressly requested in the Use and Maintenance Manual.
- Never use tools improperly or in bad conditions, e.g. pliers instead of the special spanners etc.
- Keep the maintenance area clean and dry, and immediately dry any traces of water and oil.
- The leaks of pressurised fluid through small holes are almost invisible and can be strong enough to hole your skin.
Before checking for leaks, protect your eyes with protective goggles with side visors.
Do not use your hands, but use a piece of cardboard or wood, to detect suspected leaks of pressurised liquid.
Wounds caused with pressurised fluid can cause serious infections. In this case, consult a doctor immediately.
- Do not gather oily or greasy cloths as they constitute a fire risk.
These cloths must be deposited in a closed metal container.
- Immediately replace any Danger, Attention or Instruction warning plate that is no longer readable or missing.



Attention

Never perform jobs with the removal or deposit of material (welding, drilling, sanding) without the Manufacturer's authorisation and instructions.

- At the end of maintenance or repair operations, before starting the machine, check that no tools or other material are left inside the compartments containing moving parts or close to moving parts.
- Always ensure that both the machine and relevant accessories are in good order.



Danger

- Before restarting the machine ensure that all maintenance operations have been completed correctly and that starting of the machine will not result in any risk conditions.
 - Immediately after having completed the operation, restore and check the safety devices that were removed during the maintenance or repair operation.
-



Attention

All untreated surfaces (guides, racks, etc.) should be cleaned and then lubricated.

Key to symbols used in the chapter:



Brush greasing



Inject grease through the grease nipple
Grease by means of a grease gun

Consultation of technical documents

Before performing maintenance work on the machine, read the technical documentation supplied by the Manufacturer and the suppliers of individual commercial parts of the machine.

In particular consult:

- the “operating instructions”;
- the “user instructions for the operating machine”;
- the diagrams of the electrical, hydraulic, pneumatic systems, etc.

In any case work on the machine only if you are in possession of adequate technical know-how. The Manufacturer’s technical service is at your complete disposal for any information concerning maintenance work to be carried out on the parts supplied.



Attention

In case of operating faults, do not attempt to solve any anomalous situations that may occur using makeshift means.

Spare parts

The use of non-authentic spare parts may cause machine malfunctions, which in turn may lead to hazardous situations for the operator and any individuals working near the machine.



Attention

Always use authentic spare parts.

Bucket configuration

Before performing any maintenance operation, position the bucket perfectly resting on a flat and well-lit surface.

Moreover, if the operating machine remains connected, it is necessary to:

- discharge the aux. system pressure completely and take the respective control to the “**0-auger standstill**” position;
- engage the parking brake;
- turn the engine off;
- remove the ignition key from the dashboard.

If the equipment must be disconnected from the operating machine, see “Equipment removal”.



Pollution hazard

It is prohibited to disperse old brushes, rubber or plastic components, solvents, oil and lubricants into the environment.

Collect and dispose of these components according to the provisions in force in each country.

Scheduled maintenance

Daily checks

- Check the oil level in the operating machine hydraulic tank as indicated in the respective user manual.
- Check the flexible hoses, the fittings and the other components of the hydraulic system so as to prevent breakage and pressurised oil leaks.
Eliminate any leaks in the hydraulic system.
- Check that all the guards are correctly installed and that the safety devices are efficiently operating.
- Check that the structures of the bucket and of the respective accessories are whole and show no signs of cracks or deformations.
- Check both directions of rotation of the auger.
- Wipe all unpainted parts with a cloth soaked in oil.
- Wash the bucket (see “Cleaning the beton mixer bucket”).
- Grease the bearings and inner seals.

Every 50 operating hours or every week

- Check tightness of the bolts and screws connecting parts to the bucket.
- Check tightening of fittings and flexible hoses and the piping in general.
- Lubricate the transmission chain.
- Check pinions, chains and auger for wear.
- Check efficiency of the seals inside auger supports.

Every 200 hours

- **For models 100, 150, 200, 200SL 250, 250SL, 300, 300SL, 350**
Check for the correct operation of the auger rotation bearings and replace them, if necessary.

Every year

Carry out all the checks specified on the sheet on page 78.

Maintenance operations

Cleaning the beton mixer bucket

- Empty the bucket completely.
- Position the bucket on the ground as during the aggregate loading phase and in an area where it is possible to collect and dispose of the water and additives used for washing.



Attention

Never wash the bucket with chemical degreasers: they can cause infiltrations that are dangerous for the machine.

- Stop the operating machine, engage the parking brake, remove the key from the dashboard and block the arms of the same to prevent their accidental downstroke.
- Wash the bucket with strong jets of water, paying particular attention to:
 - mixing zone;
 - unload outlet;
 - auger shaft supports.



Attention

If necessary, it is possible to open the protection grate. Carefully check what is described in chapter "Opening the protection grate".



Pollution hazard

Do not disperse dirty water or with additives into the ground, but collect water and dirt in a relevant tank in order to re-cycle them in compliance with the provisions of the applicable standards.

- Wipe all unpainted parts with a cloth soaked in oil.

Lubrication

Preliminary information

The bucket operates with the working machine hydraulic system, therefore to top-up or replace the hydraulic oil, consult the machine use and maintenance manual.

The bucket has been designed reducing any maintenance and lubrication intervention to minimum.



Pollution hazard

It is prohibited to disperse waste oil into the environment.

Place oily cloths, lubricants, solvents and filtering cartridges inside suitable containers and dispose of them in compliance with the provisions and Standards in force in each single country.



Attention

Do not mix different types of oil, restore the level of lubricants exclusively with lubricants of the same type as those contained in the respective tanks.

Only use the lubricants recommended by the Manufacturer or the corresponding lubricants indicated in the relative table.



Danger

Only use the greases indicated.

Other products may be incompatible with the product used by the manufacturer for the first greasing.

Incompatible greases that are mixed or used at a second stage can develop substances affecting machine operation with potentially serious safety consequences.

Lubricating grease comparative table

	TOTAL	MOBIL	ESSO	AGIP	IP	BP
Grease 	HERELDA 2	FARM T.GREASE	CAZAR K2	GREASE 16	AUTO GR CH	GREASE A
Grease 	MULTIS EP 2	MOBILUX EP 2	BEACON EP 2	GR MU EP 2	ARTHESIA GR EP 2	ENERGREASE NM EP 2
Protective oil	OSYRIS ACR	MOBILARMA 246	RUSBAN 398	RUSTIA 82	IDEX FLUID PM GR 2	C.P.F. 21

Lubricating points



Danger

This operation must be performed with the operating machine engine at a standstill, parking brake engaged, hydraulic system disabled and key removed from the dashboard. Limb amputation hazard.

Remove the protection casing (1) to grease the transmission chain (2).

Brush a thin layer of grease onto the chain (see "Lubricating grease comparative table").

After lubrication, reposition the protection casing (1).



Attention

In heavy duty work and environmental conditions, shorten the frequency of the lubrication intervals.

Supply grease nipples

Top-up the greasers present on the bucket as follows:

- Always clean the greasing heads to prevent the infiltration of dirt.
- Feed all greasers using a standard grease pump.
Fill the type of grease described in the "Lubricants correspondence table".
Clean the excess waste grease using a cloth.



Attention

In heavy duty work and environmental conditions, shorten the frequency of the lubrication intervals.

- In **100, 150, 200, 200 SL, 250, 250 SL, 300, 300 SL and 350 models**, bearings are water-tight with permanent lubrication and do not require topping-up during the first 200 operating hours of the machine. Once this time has elapsed, it is necessary to make sure that bearings are working correctly.
- In **450, 600, 750, 1000 models**, grease the bearings and the inner seals until the old grease is completely drained.
Clean the waste grease using a cloth.
Repeat greasing leaving a part of the new escaping grease.

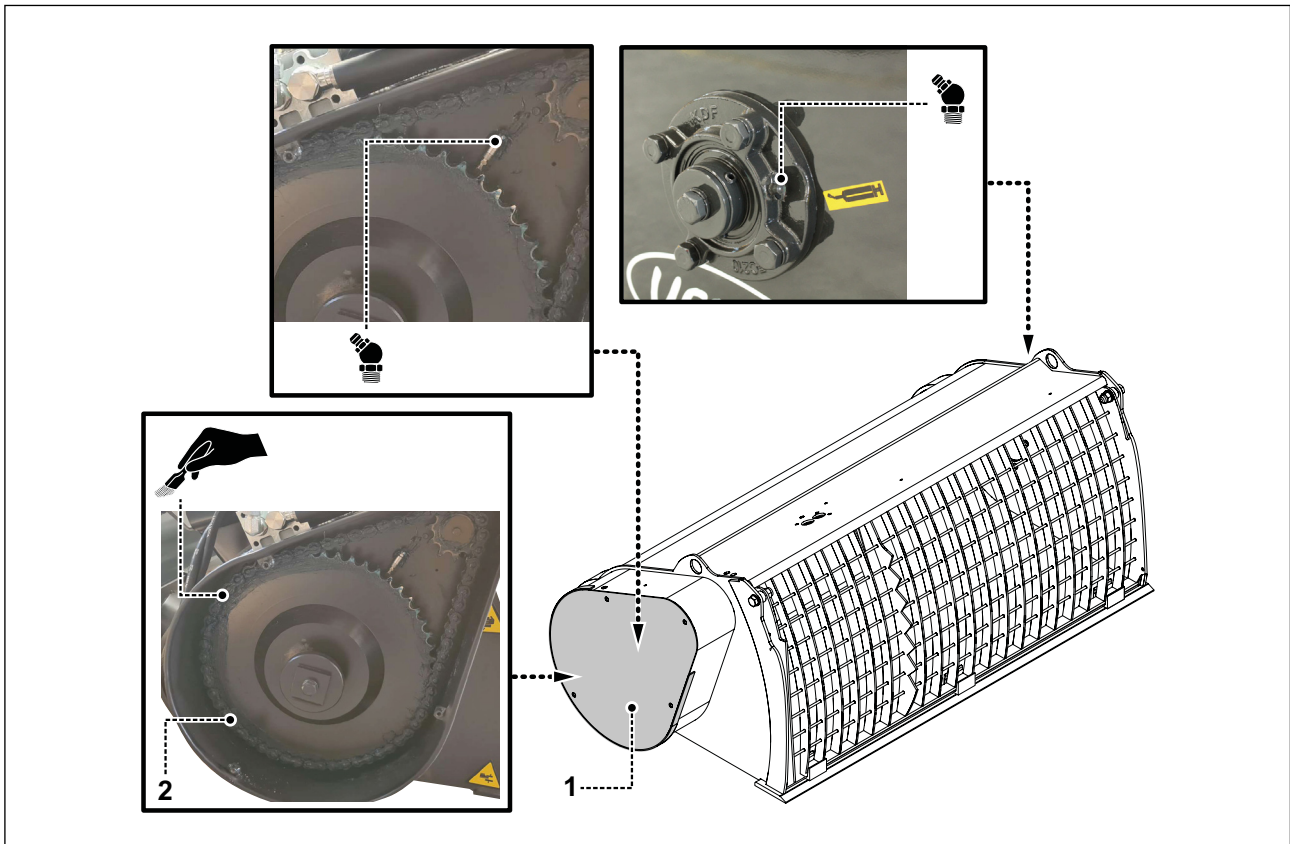


Attention

Grease the bearings and the inner seals with the auger in movement. Keep it moving strictly for the time required for greasing.

M - MAINTENANCE

Greasing points diagram (Models 100-150-200-200SL-250-250SL-300-300SL-350)



Attention

As mentioned above, the auger rotation bearings are hermetically lubricated and do not require maintenance during the first 200 operating hours of the machine. Once this time has elapsed, it will be necessary to make sure that they work properly and, if necessary, replace them with new bearings having the same characteristics or lubricate them. In case of lubrication it is **MANDATORY** to use a manual pump and take care to stop lubrication as resistance to pump pressure increases. Any grease leakage means that seals are damaged and, therefore, involved component must be immediately replaced.

Lubrication interval

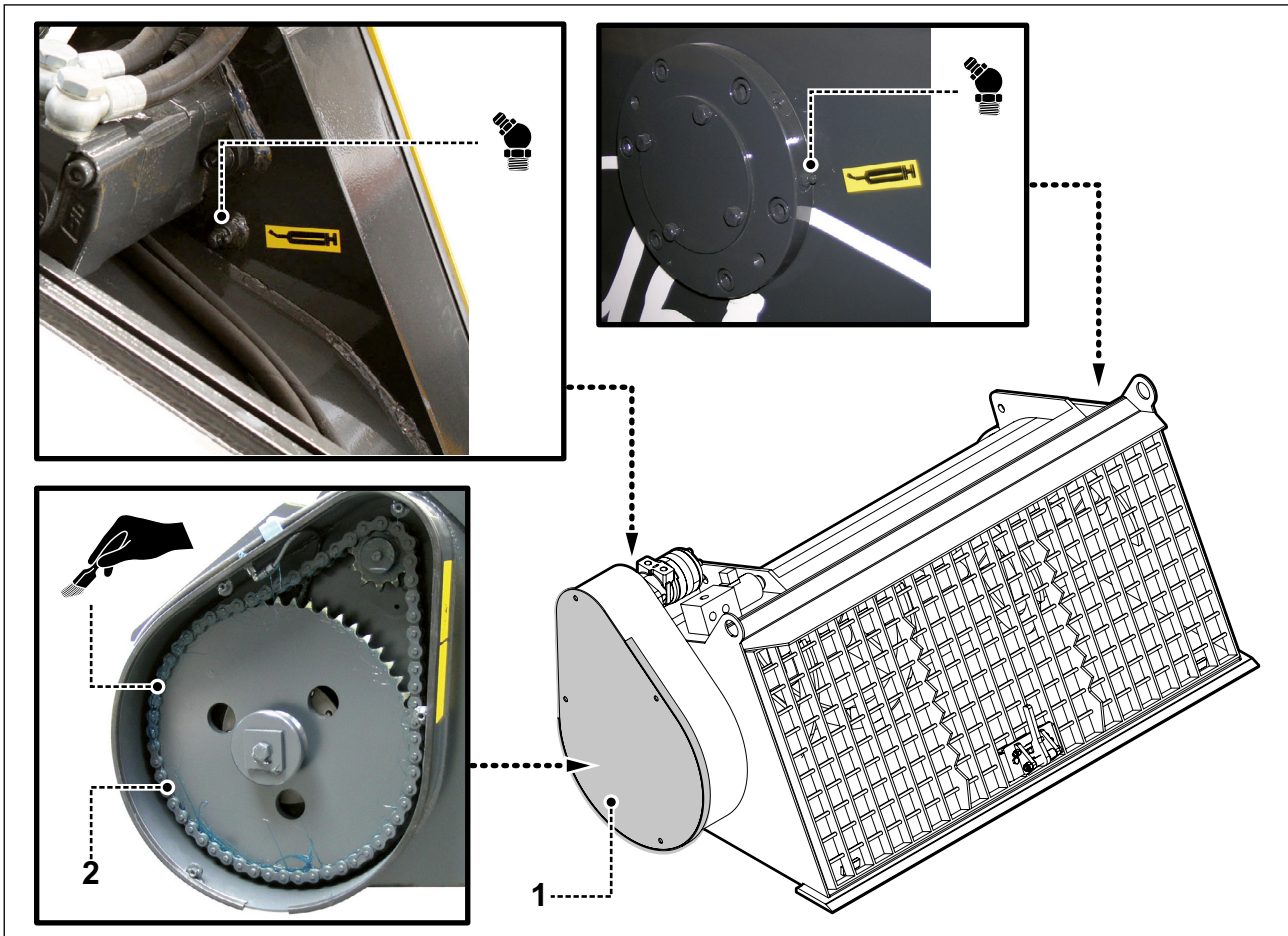
Temperature	Working condition			Temperature	Grease type
	Clean	Dusty	Dusty and damp		
Up to 50°C	12 months	6 months	4 months	Standard	Lithium
Up to 70°C	12 months	4 months	1 month		
Up to 100°C	6 months	2 months	15 days		



Note

The values shown in the table are referred to 8-10 hour operation per day.

Greasing points diagram (Models 450-600-750-1000)



Tightening torque table

Unless differently indicated, tighten the screws present on the mixer bucket using the tightening torque values (Nm) indicated in the following table.

Dimensions of the screws	Tightening torques (Nm)		
	Class		
	8.8	10.9	12.9
M4	2.7	3.8	4.6
M5	5.5	8.0	9.5
M6	9.5	13.0	16.0
M8	23.0	32.0	39.0
M10	46.0	64.0	77.0
M12	80.0	110.0	135.0
M14	125.0	180.0	215.0
M 16	195.0	275.0	330.0
M18	270.0	390.0	455.0
M20	385.0	540.0	650.0
M22	510.0	720.0	870.0
M24	660.0	930.0	1100.0
M27	980.0	1400.0	1650.0
M30	1350.0	1850.0	2250.0

Ordering spare parts

Consult the specific spare parts catalogue when you need to order spare parts.

Spare parts must be ordered to the retailer or assistance centre and the orders must include the following indications:

- Type and model of equipment.
- Part number of the requested part.
If you do not have this number, list the number of the table in which the corresponding reference is provided.
- Name of the part and desired quantity.
- Preferred transportation means.
If this item is not specified, the retailer or assistance centre, though dedicating special care to this service, will not respond to shipping delays due to force majeure.
Shipping costs are always charged to the addressee.

Placing the machine out of service for a prolonged period of disuse

In the event of prolonged disuse perform the following operations on the machine:

- Thorough cleaning.
- Check that the protection grate is fixed correctly.
- Place the bucket on wooden pallets with suitable dimensions to guarantee the stability of the equipment.
- Lubrication of all moving parts.
- Anti-rust surface treatment on all unpainted metal parts (apply oil or MoS2 spray).
- Cover the machine with a waterproof tarpaulin to protect it from dust and damp.
- Store in a dry and protected place with access limited to authorised persons only.

Service life

The actual service life of the equipment, if all the checks, the envisioned maintenance operations and inspections are carried out, is 10 years from its first start up. After this period of time the use of the equipment is prohibited if not overhauled and checked by the Manufacturer.

Further overhauls must be carried out every 2 years.

Control register

Storage instructions

This Control register must be considered as a part of the equipment and must accompany it during its whole life until its final dismantling.

Instructions for the drawing up

These instructions are supplied according to the provisions known when the equipment was first placed on the market.

The register is arranged to take note of the following events regarding the useful life of the equipment according to the layouts proposed:

- Transfers of ownership
- Replacing mechanisms, structural elements, safety devices and relative components.
- Serious failures and their repairs.
- Maintenance and periodical checks.



Note

Whether the sheets of the present register were not enough, add the necessary sheets, drawn according to the schedules here indicated.

The user must include the identification data of the equipment on the additional sheets.

These sheets will be an integral part of the present register.

Authorised persons

This documentation must be drawn up by the owner of the equipment, or by someone directly delegated by him, as must the tests to be carried out by specialised personnel; verification calculations (stability, secondary frame, etc.) must be carried out by legally qualified persons.

Storage of the control register

This Register, summarising the essential technical features and data of the equipment, must be kept for the entire working life of the equipment.

The subsequent checks to be carried out within the legally envisioned times and methods and the execution of the servicing checks must also be recorded on it, as well as the inspections before commissioning.

The inspections and any extraordinary checks to be carried out in case of constructive changes, structural repairs or use change compared to that established by the manufacturer must also be recorded.

Identification of equipment

Equipment model:

Serial number and year of manufacture:

Type and frame no. of operating machine:

Manufacturer information: **U.EMME s.r.l.**

Via dell'artigianato 19 - 47015 Modigliana (FC)

Tel. +39 0546 941725 - Fax +39 0546 940050

e-mail: info@uemme.com

www.uemme.com



Note

For all technical data of the equipment and operating instructions, refer to the "User and maintenance manual" this register is attached to.

OPERATING MACHINE

- Manufacturer:

- Frame Number:

- Plate:

EQUIPMENT:

- Manufacturer:

- Type: Year of manufacture:

- Date of commissioning:

- Serial No.:

- CE Declaration of Conformity:

INSTALLED SAFETY DEVICES

☐ YES

.....

.....

.....

☐ NO

OPERATING MACHINE CONTROL STATION

☐

M - MAINTENANCE

STRESS CHECK

☐ YES

☐ NO

ATTACHMENTS

☐ User and maintenance manual:

☐ Control register:

☐:

☐:

Place *Date*

M - MAINTENANCE

Equipment delivery to first owner

The equipment and optionals listed below have been delivered from the Company U.Emme to the Company:

.....

according to the conditions established in the contract.

Machine	Serial No.	Year of manufacture

Date.....

M - MAINTENANCE

Transferring the property

On

The ownership of the concerned equipment is transferred to the Firm/Company:

.....
.....
.....

We certify that, on the above-written date, the technical, dimensional and functional features of the concerned equipment comply with those intended originally and that any variations have been transcribed in this Register.

Seller

Buyer

.....

.....

On

The ownership of the concerned equipment is transferred to the Firm/Company:

.....
.....
.....

We certify that, on the above-written date, the technical, dimensional and functional features of the concerned equipment comply with those intended originally and that any variations have been transcribed in this Register.

Seller

Buyer

.....

.....

On

The ownership of the concerned equipment is transferred to the Firm/Company:

.....
.....
.....

We certify that, on the above-written date, the technical, dimensional and functional features of the concerned equipment comply with those intended originally and that any variations have been transcribed in this Register.

Seller

Buyer

.....

.....

M - MAINTENANCE

Mechanism replacement

Date: Replaced element
Manufacturer Supplier
Reason for the replacement:
.....
.....

The representative of the company responsible for the replacement The user
.....

Date: Replaced element
Manufacturer Supplier
Reason for the replacement:
.....
.....

The representative of the company responsible for the replacement The user
.....

Structural elements replacement

Date: Replaced element
Manufacturer Supplier
Reason for the replacement:
.....
.....

The representative of the company responsible for the replacement The user
.....

Date: Replaced element
Manufacturer Supplier
Reason for the replacement:
.....
.....

The representative of the company responsible for the replacement The user
.....

M - MAINTENANCE

Safety device and relevant component replacement

Date: Replaced element

Manufacturer Supplier

Reason for the replacement:

.....

.....

The representative of the company responsible for the replacement The user

.....

Date: Replaced element

Manufacturer Supplier

Reason for the replacement:

.....

.....

The representative of the company responsible for the replacement The user

.....

Serious failures and their repairs

Failure description:

.....

Causes:

.....

Performed repair:

.....

.....

Place and Date:

The representative of the company responsible for the replacement The user

.....

Failure description:

.....

Causes:

.....

Performed repair:

.....

.....

Place and Date:

The representative of the company responsible for the replacement The user

.....

Periodical inspections

The user must respect the maintenance and surveillance programme described in this user's manual.

The person responsible for the equipment must report, in the following pages, the check and maintenance operations for the periodical inspections on the equipment.

The check must be carried out in relation to the use intensity of the equipment and the particular work environment.

All routine and extraordinary maintenance operations must be transcribed in the following servicing checks, specifying the operation carried out, the date, the work hours and who carried them out (operator, qualified workshop, manufacturer).

The equipment must be taken for a check at an Authorised workshop at least once a year.

The correct use of the equipment is the sole responsibility of the final user, responsible for choosing the product for dimensions and capacities, being the dealer unable to guarantee that the equipment is suitable for its effective use without knowing the same that, however, must fall within the use limits envisioned by the use and maintenance manual.

For maintenance operations, please refer to section M - maintenance of the equipment.

M - MAINTENANCE

Inspections

General

In order to assure the safe functioning of the equipment, the correct working and functioning conditions must be maintained. Therefore a regular check is necessary.

Inspections must be scheduled by the user.

Inspection before use

Before use the operator must check the equipment.

Intervals between inspections

Depending on the duration and operating conditions and work place, the equipment must be inspected how and when required, but at least once a year.

Description of action	Executor	Operating hours	Date	Signature

M - MAINTENANCE

EVERY YEAR

- ☐ **IDENTIFICATION PLATES**
- ☐ **CAPACITY PLATES**
- ☐ **LOADLESS OPERATING TESTS**
- ☐ **OPERATING TESTS AT PLATE LOAD VALUES**
- ☐ **WEAR, CLEARANCES**
- ☐ **DEFORMATIONS**

Various:
.....
.....

Comments and notes:
.....
.....

Date.....

Authorised workshop

The machine manager

.....

.....

M - MAINTENANCE

Forms for periodical inspections

Comments:

.....

Name / Company of Inspector:

.....

.....

The periodical inspection **has** / **has not** been carried out.

Faults **have** / **have not** been detected: see test result (*)

.....

.....

Date..... Signature.....

The periodical inspection **has** / **has not** been carried out.

Faults **have** / **have not** been detected: see test result (*)

.....

.....

Date..... Signature.....

The periodical inspection **has** / **has not** been carried out.

Faults **have** / **have not** been detected: see test result (*)

.....

.....

Date..... Signature.....

The periodical inspection **has** / **has not** been carried out.

Faults **have** / **have not** been detected: see test result (*)

.....

.....

Date..... Signature.....

(*) Delete unwanted part.

Warning

The equipment does not require particular attention for disposal because more than 90% (in weight) is made up from materials that can be re-cycled.

Scrapping must be performed using safety measures taking into account the logistic, environmental and wear conditions of the equipment itself.

Nonetheless, follow the general rules below:

- Wear protective clothing and accessories (helmet, safety footwear, gloves, safety goggles and face mask if necessary) approved in accordance with the prevailing accident-prevention standards.
- Disconnect the machine from all energy sources.
- Use suitable lifting means as indicated in the "Transport" section of the "Lifting systems" chapter.



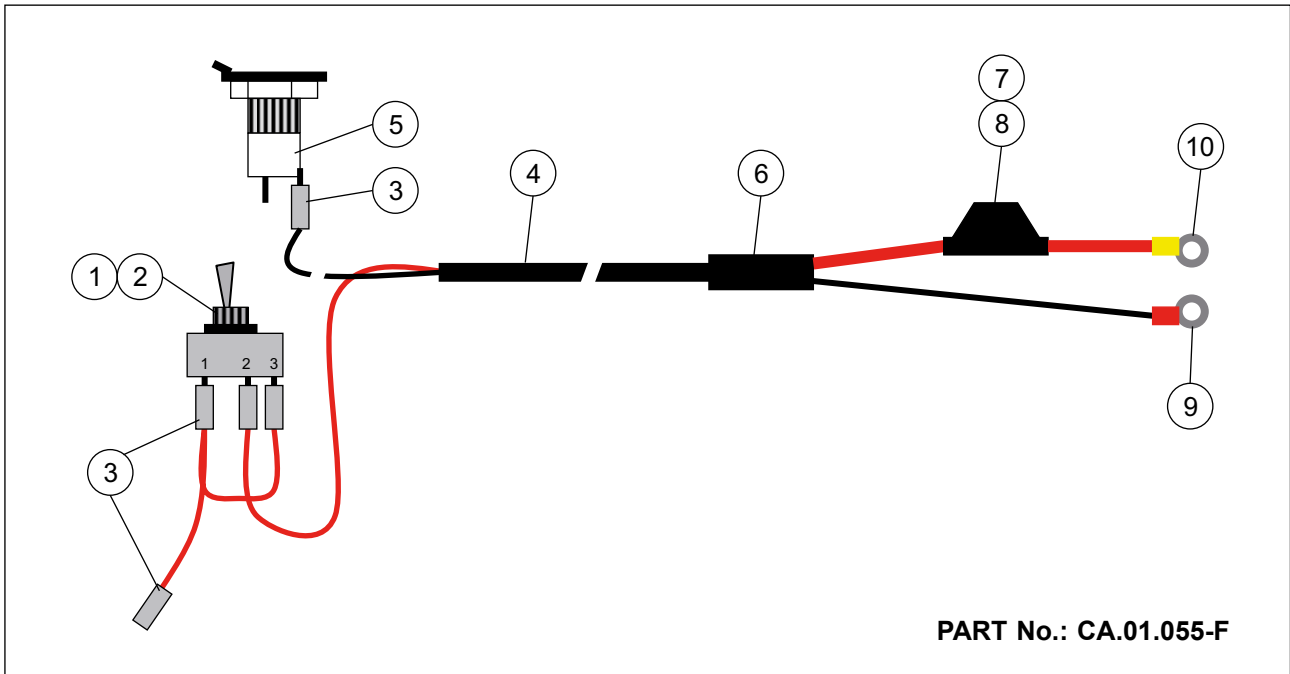
Attention

The machinery must be demolished and disposed of by specialized, qualified technicians in accordance with all rules on the scrapping of industrial products.

O - ATTACHMENTS

Wiring diagram

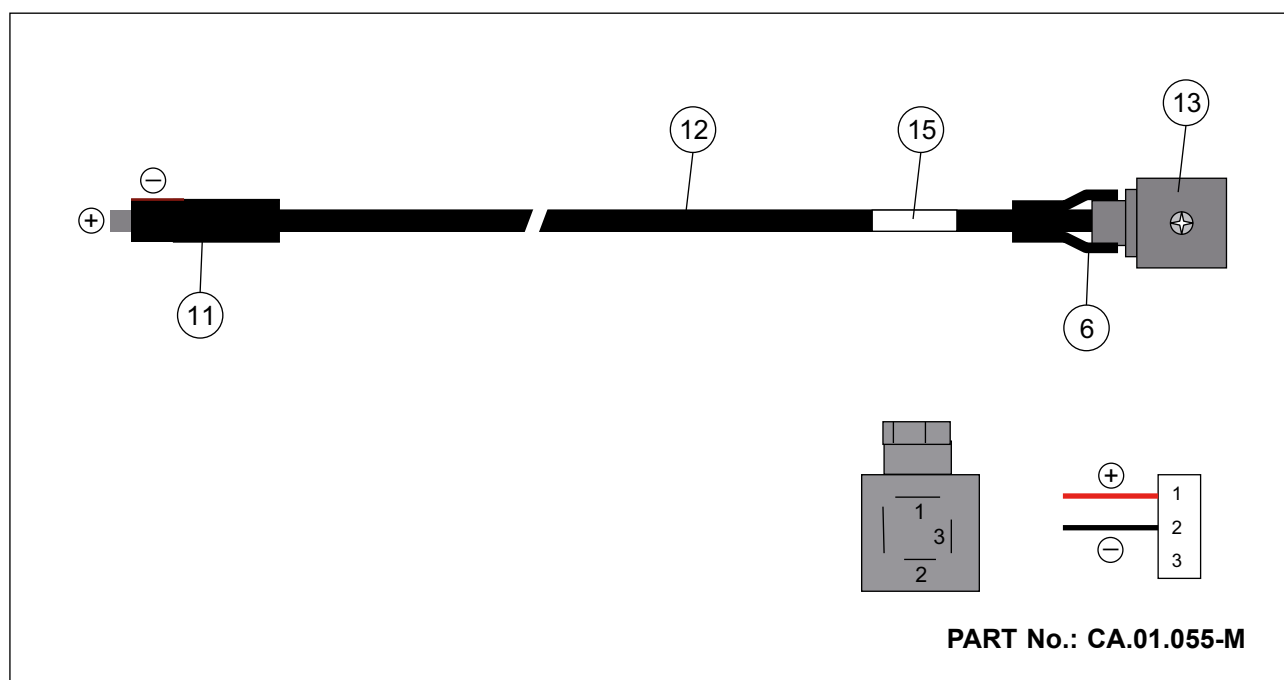
Wiring kit for machines NOT set-up for electric connection



Ref.	Description
1	Diverter (ON stable - OFF- ON with spring return)
2	Semi protection
3	White insulated Fast-ON
4	Red/black cable 2x1 mm ² with black PVC jacket
5	Cigarette lighter from panel
6	Heat shrinking with HOT-MELT
7	Fuse holder with red co-pressed cable
8	Fuse 10A
9	Terminal
10	Terminal

O - ATTACHMENTS

Wiring kit for machines set-up for electric connection



Ref.	Description
11	Pin for cigarette lighter
12	Red/black cable 2x1 mm ² with black PVC jacket
13	DIN connector for solenoid valve with standard fitting
6	Heat shrinking with HOT-MELT
15	Label with CA.01.055 marking, code, week and year

[illegible]

[illegible]



U.EMME s.r.l.

Via dell'artigianato 19 - 47015 Modigliana (FC)

Tel.: +39 0546 941725 - Fax: +39 0546 940050

e-mail: info@uemme.com

www.uemme.com